Introduction :

The design of electronic circuits using discrete components is effective for small circuits. When circuit complexity is more, more number of electronic components are required with more of connections blw them.

Designing of such circuits may be critical, time consuming, less reliable, utilise more power consumption & system performance decreases.

To avoid these problems, all the components with their inter connections are fabricated on single chip. such a circuit is called Integrated circuits.

Advantages of IC's :

1. Complexity of circuit is reduced
2. Cost is reduced.
3. Increase in system reliability.
4. System performance is improved.
5. Speed is increased
6. Power consumption is reduced.

Classification of IC's :

According to mode of operation, integrated circuits are classified into 2 types.

1. Linear integrated circuits
2. Digital integrated circuits

1. Linear IC's :

Linear IC's are equivalent to transistor circuits such as amplifiers, filters, modulators, frequency multipliers & so on..

→ Linear Ic's will operates on the continous signals.
→ Most commonly used linear Ic is Op-amp.

2. Digital Ic's :

Digital Ic's are used to form circuits which have 2 levels of voltage or current i.e., high & low.
→ Digital Ic's will operates on 0 & 1 inputs.
→ The ~~circuits~~ examples of digital Ic's are logic gates, counters, shift registers, multiplexers & flipflops.

The above two Ic's (linear & digital) are developed by using monolithic & hybrid technologies.

Monolithic Ic's :

→ In this Ic's all circuit components both active & passive elements & their interconnection are manufacted into a single chip.
→ Monolithic Ic's provide good thermal stability because all the components are integrated on the same chip.
→ Monolithic Ic's are used in now-a-days in different applications

Hybrid Ic's :

→ In this Ic's separate component parts are attached to a ceramic substrate & inter connected by wires.
→ This technology is more adoptable to small quantity custom circuits.
→ Active components such as transistors & diodes as well as monolithic Ic's are then connected to form a complete circuit.

Chip size and Circuit Complexity :

→ The electronic device technology was dominated by the vaccum tubes.

→ The present day electronics is the result of the invention of a transistor in 1947.

→ The concept of Ic was introduced at the beginning of 1960 by both texas instruments & fair child semiconductors.

→ since that time a size & complexity of Ic's have increased rapidly.

We can classify the Ic's using levels of integration.

1. small scale integration (SSI)
2. Medium scale integration (MSI)
3. Large scale integration (LSI)
4. Very large scale integration (VLSI)
5. Ultra large scale integration (ULSI)

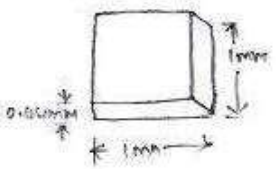
1. SSI :

→ The first Ic was implemented in this level. These Ic's contains 3-30 logic gates per chip (or) 100 transistors ~~for~~ Per chip.

→ These Ic's are used in logic gates & flipflop applications.

→ These chip size is very small.

→ These Ic's are available in 14 (or) 16 pin DIP packages.

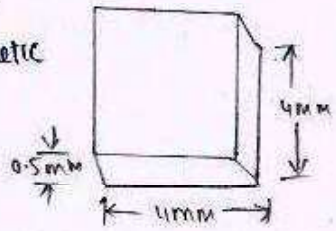


→ The next large commerical available Ic's are called MSI. It

standa contains 30 to 300 gates for chip or 100 to 1000 transistors.

These circuits used in encoders, decoders, counters multipliers, demultiplexer, multiplexer, register, arithmetic circuit applications

These are available in 24 pin DIP.

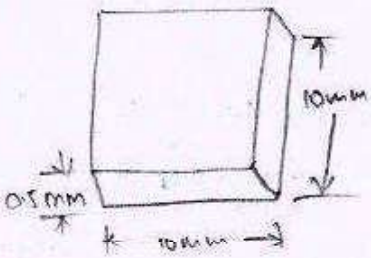


Large Scale Integration:

→ LSI means large scale integration. These IC's are still bigger than MSI. It contains 300 to 3000 gates per chip or 1000 to 2000 transistor per chip. These IC's are used in µ microprocessor & memories.

Very Large Scale Integration:

→ VLSI IC's are fabricated with more than 3000 gates per chip (or) 2000 to 1 lakh transistors per chip. These IC's are used in advance microprocess i.e., 16 bit & 32 bit.

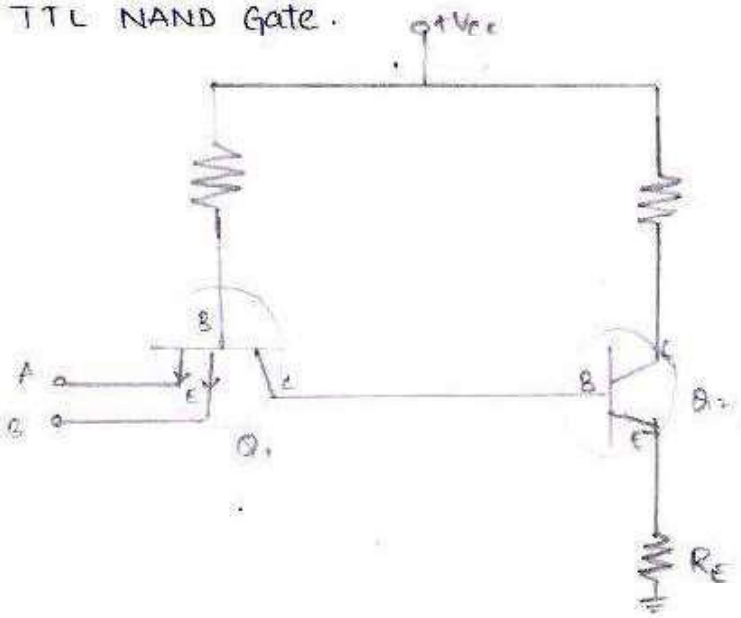


ULSI:

ULSI means Ultra Large scale integration. These IC's are advance IC's it contains more than VLSI. These IC's are used in 64 bit microprocessor applications.

Standard TTL circuit & characteristics :

TTL means Transistor Transistor logic that means we are able to design any logic function using transistors. Here we can take a 2 i/p TTL NAND Gate.

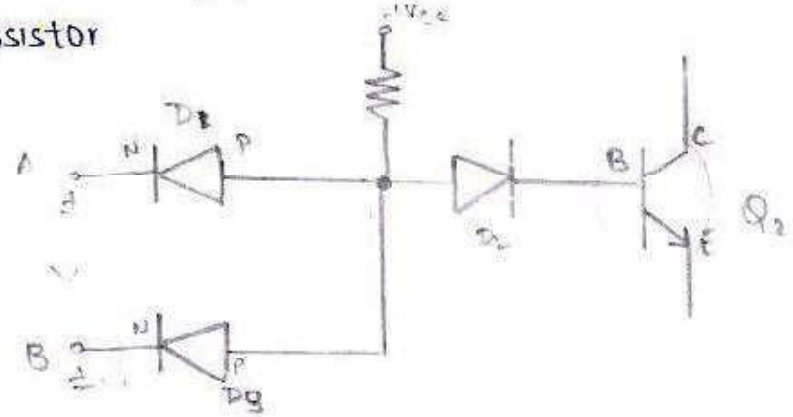


A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

→ The above fig. shows the 2 i/p TTL NAND Gate.

→ It consists of Q_1 and Q_2 transistors. It shows that Q_1 is NPN transistor having 2 emitters one for each i/p to the gate.

→ The below fig. shows the Diode equivalent circuit for Q_1 transistor



→ Here diodes D_1, D_3 represents the emitter base junction of Q_1 & D_2 represents the collector base junction of Q_1.

→ If i/p voltages are grounded it represent Low (or) if i/p voltages A & B are connected to +5V supply, it represents high.

→ If either A (or) B (or) both i/p's are low the corresponding diodes conducts due to this Q_1 is going to ON. So the total voltage flows into the ground.

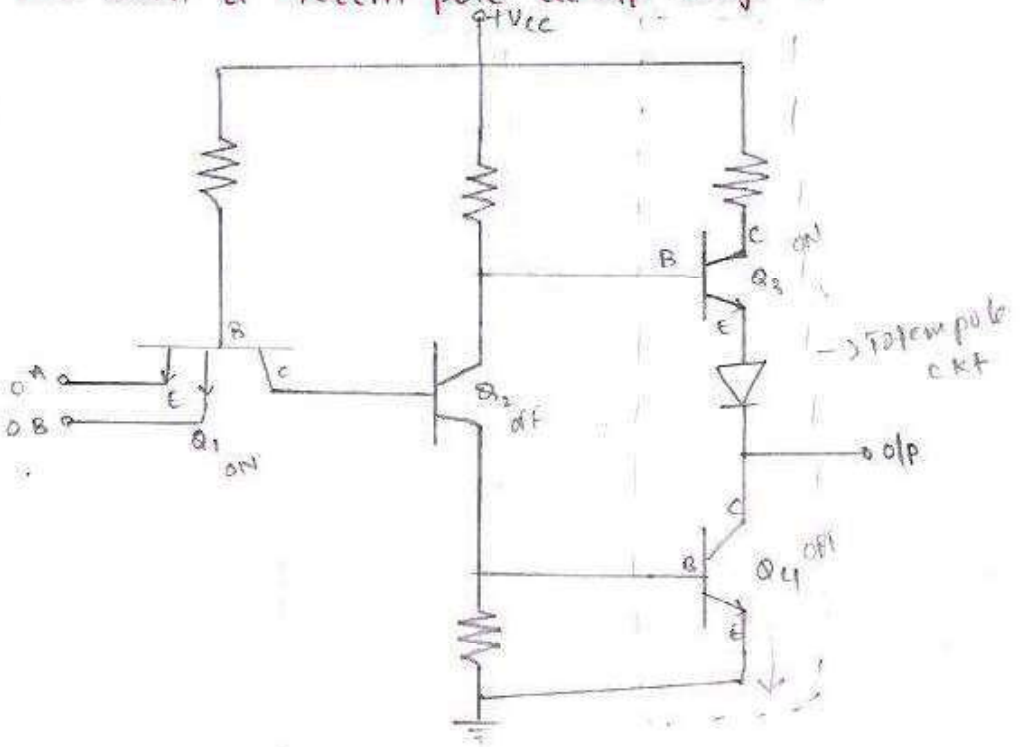
→ This reduces the base voltage of Q_2 is almost zero. Therefore Q_2 transistor is going to OFF & then o/p is high.

→ When both i/p's are high, the corresponding diodes are not conducting. So i/p of Q_1 is OFF. So no current flows to ground. The total voltage flows into the base of Q_2. So Q_2 is going to ON. So the total current flows into the grounded through Q_2 transistor. So o/p is low.

Disadvantage :

→ The main disadvantage of TTL NAND gate is that o/p resistance is increased at o/p logical high. Due to this less number of i/p's that can be connected into the another circuit. This drawback can over come by Totem pole circuit used at o/p stage.

TTL with a Totem pole at o/p stage :



→ To solve the problem with high o/p resistance of single o/p stage by using Totem pole (push pull) o/p.

→ It consists of 2 NPN transistors Q_3 & Q_4 and diode D.

→ Q_3 transistor is called as Pull-up transistor and Q_4 transistor is called Pull-down transistor.

→ Transistors Q_3 and Q_4 forms totem pole configuration.

Operation :

→ When A & B are low either A or B are high Q_1 is OFF ON and Q_2 is OFF.

→ ~~Due~~ to Q_2 is OFF due to this Q_4 is OFF & Q_3 is ON. So o/p is high (or) logical one.

→ When both i/p's are high Q_1 is OFF and Q_2 is ON due to this Q_4 is ON and Q_3 is OFF. So o/p is low.

→ Here the diode D is used to isolate the o/p from supply voltage.

→ The main advantage of TTL with Totem pole o/p stage is that low o/p resistance at Logical One (or) high.

→ Because when Q_3 is ON it acts as an emitter follower.

Analysis :

NOTE : In TTL NAND Gate diode D is connected b/w pull up (Q_3) & pull-down (Q_4) transistors. The purpose of diode is to isolate the o/p node from supply voltage. Whether it is true (or) false.

Ans :- True

Explanation :

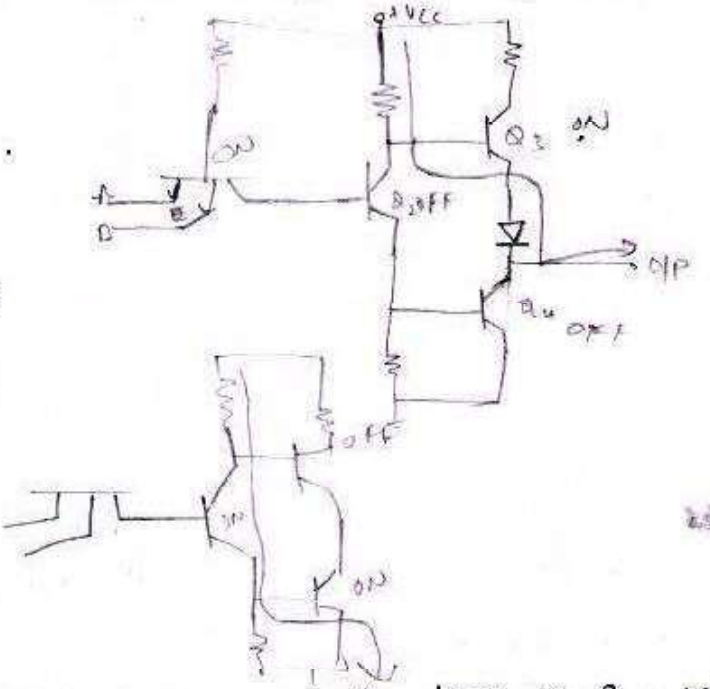
→ The required voltage is 0.9 V to turn ON the transistor.

→ We can analyse the circuit under 2 conditions.

Correct

Condition 1 :- When AB = 0 (or) either A (or) B = 1.

Due to that giving i/p's Q_1 = ON ; Q_2 = OFF ; Q_3 = ON ; Q_4 = OFF and o/p Y = 1.



Condition 2 :- When AB = 1

Due to that Q_1 = OFF ;

Q_2 = ON ;

Q_4 = ON ;

Q_3 = OFF ;

So o/p is low.

→ To make Q_3 is ON, it requires 0.9 V at the base of Q_3. when diode D is not present. i.e., $V = V_{BE_3} + V_{CE_4}(sat)$

$= 0.7 + 0.2$

$V = 0.9\ V$

→ So due to this diode is not present there is chance to turn ON the transistor Q_3 at o/p is low. To prevent this we, diode is connected in b/w the Q_3 & Q_4.

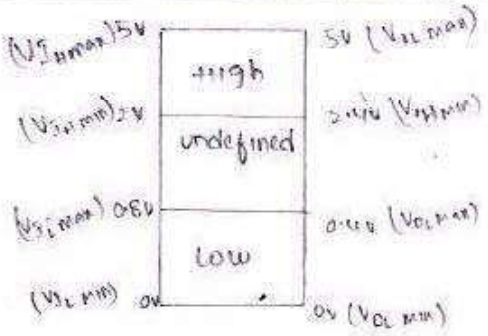
$V = V_{BE_3} + V_D + V_{CE_4}(sat)$

$= 0.7 + 0.7 + 0.2$

$= 1.6\ V$

So 1.6 V is required to turn ON the Q_3 when diode is present. But at the base of Q_3 0.9 V only appeared. So Q_3 is always going to be OFF when A & B i/p's are high.

Voltage Levels :-



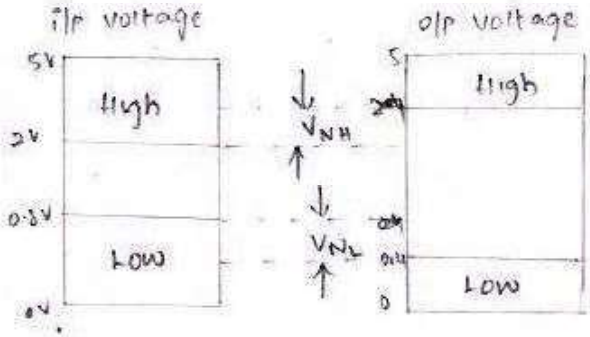
Where 1. $V_{IL\,max}$ is maximum i/p voltage in the low state is 0.8 V.

2. $V_{IH\,min}$ is minimum i/p voltage in the high state is 2 V.

3. $V_{OL\,max}$ is maximum o/p voltage in the low state is 0.4 V.

4. $V_{OH\,min}$ is minimum o/p voltage in the high state is 2.4 V.

Noise Margin :-



→ Noise margins are defined as the i/p & o/p voltages.

→ We have 2 types of noise margins.

1. High state noise margin (V_{NH})
2. Low state noise margin (V_{NL})

1. High State Noise Margin :

It is defined as the difference blw min. o/p voltage in high state and min. i/p voltage in the high state. It is given by

$$V_{NH} = V_{OH\,min} - V_{IH\,min}$$

$$= 2.4\,V - 2V$$

$$\boxed{V_{NH} = 0.4\,V}$$

2. Low state Noise Margin :

It is defined as the difference blw max. ~~o/p~~ i/p voltage

in the low state and max. o/p voltage in the low state. It is given by

$$V_{NL} = V_{IL\,max} - V_{OL\,max}$$

$$= 0.8V - 0.4V$$

$$\boxed{V_{NL} = 0.4V.}$$

Input and Output Currents (Sourcing & sinking Currents):

We have seen that TTL o/p stage is a Totem pole. It consist of 2 transistors Q_3 and Q_4.

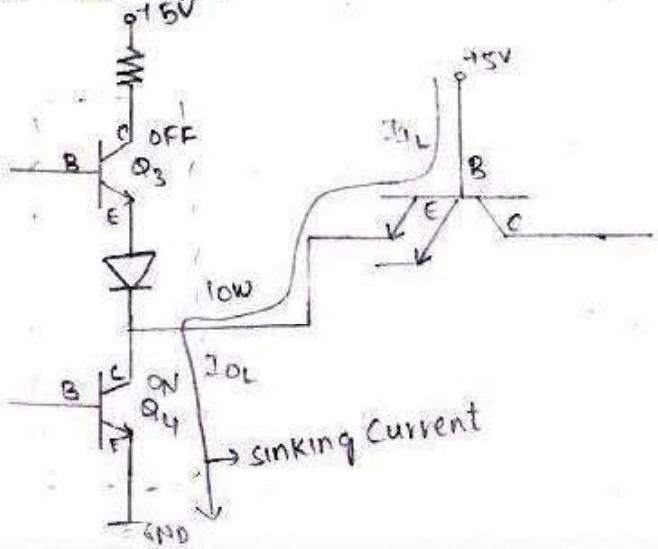
→ When Q_3 is ON, the o/p is high & it supplies current I_{IH} to drive the load. Hence Q_3 is called a Pull-up transistor.

→ When Q_4 is ON, the o/p is low & it draws current from the load to pull the load voltage down. Hence Q_4 is called pull-down transistor.

→ Since current flows out of the Totem pole when the o/p is high is called as sourcing Current.

→ When the o/p is low, current flows into Q_4 transistor, this is called as sinking Current.

→ The driver o/p is low, Q_4 sinks current from the Q_1 transistor. This current is approximately 1.6 mA & is designed I_{OL} at the o/p & I_{IL} at the i/p.



→ The driver o/p is high & Q_3 transistor ON, the current coming from source to the load.

→ It is small leakage current supplied to the reverse biased emitter base junction of the i/p. transistor of the load. It is approximately 40 μA.

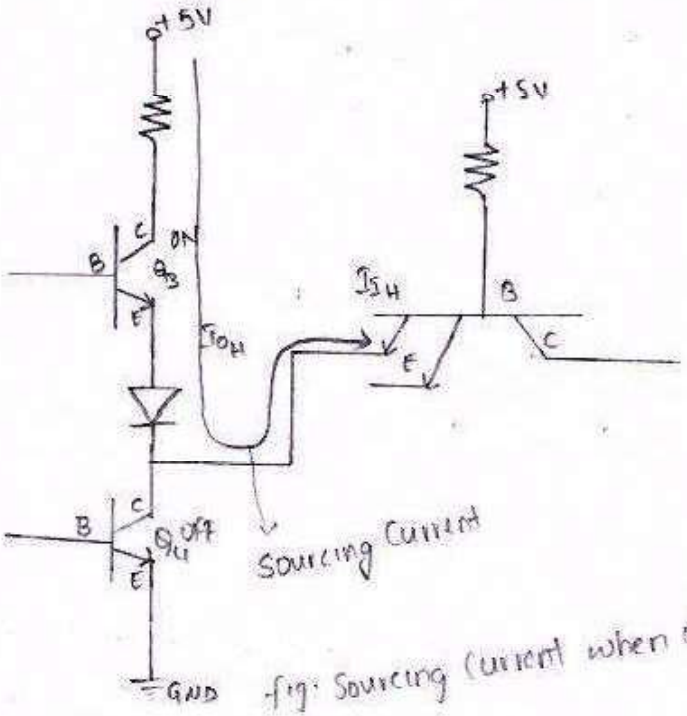


fig: Sourcing current when o/p is high

Fan-Out :

It is defined as the maximum no. of loads that a logic gate can drive. Now let us see the what will be the Fan-out when o/p is high & o/p is Low.

when o/p is high :

$$\text{Fan-Out} = \frac{I_{OH(max)}}{I_{IH(max)}}$$

For standard TTL $I_{OH(max)} = 400\,\mu A$

$I_{IH(max)} = 40\,\mu A$

$$\text{Fan-Out} = \frac{400}{40}$$

Fan-Out = 10

→ So we can say that a standard TTL O/P can typically drive 10 standard TTL I/P's.

When O/P is low :

$$\text{Fan-Out} = \frac{I_{OL}(max)}{I_{IL}(max)}$$

For standard TTL $I_{OL}(max) = 16\ mA$

$I_{OH}(max) = 1.6\ mA$

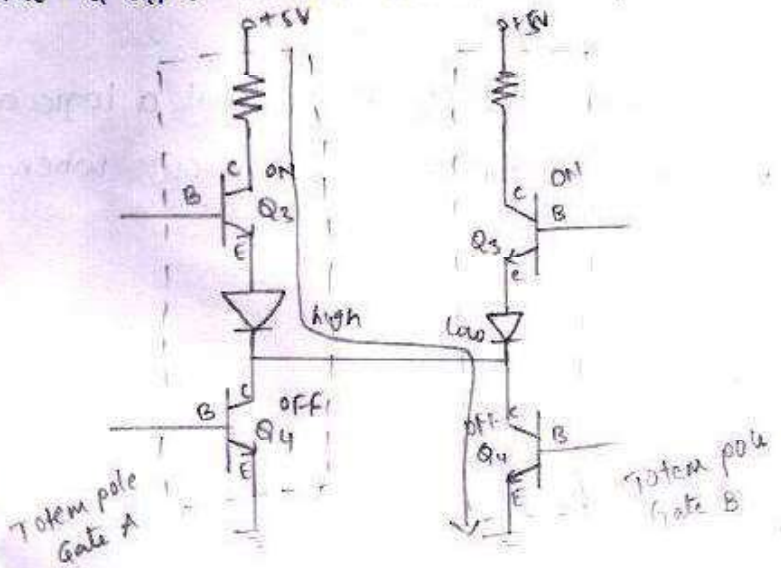
$$\text{Fan-out} = \frac{16}{1.6}$$

Fan-Out = 10

→ So we can say that a standard TTL I/P can typically drive the standard TTL O/P.

Open Collector O/P's :

We have seen Totem pole O/P. one problem with Totem pole O/P is that 2 O/P's cannot be tied together. It is shown in below fig.



→ When the totem pole O/P's of 2 separate gates are connected together.

→ Suppose the O/P of gate-A is high due to that Q_{3A} is ON & Q_{4A} is B Off.

→ Similarly the o/p of gate B is low due to that Q_{3B} is OFF & Q_{4B} is ON.

→ In this situation Q_{4B} acts as a load for Q_{3A}.

→ since Q_{4B} is a low resistance load because it draws high current around 55 mA.

→ This current may not damage Q_{3A} transistor (or) Q_{4B} transistor immediately with in a short period due to this performance of the ckt is decreased.

This drawback can overcome by TTL open collector o/p. The below fig. shows the two i/p NAND gate with an open collector o/p.

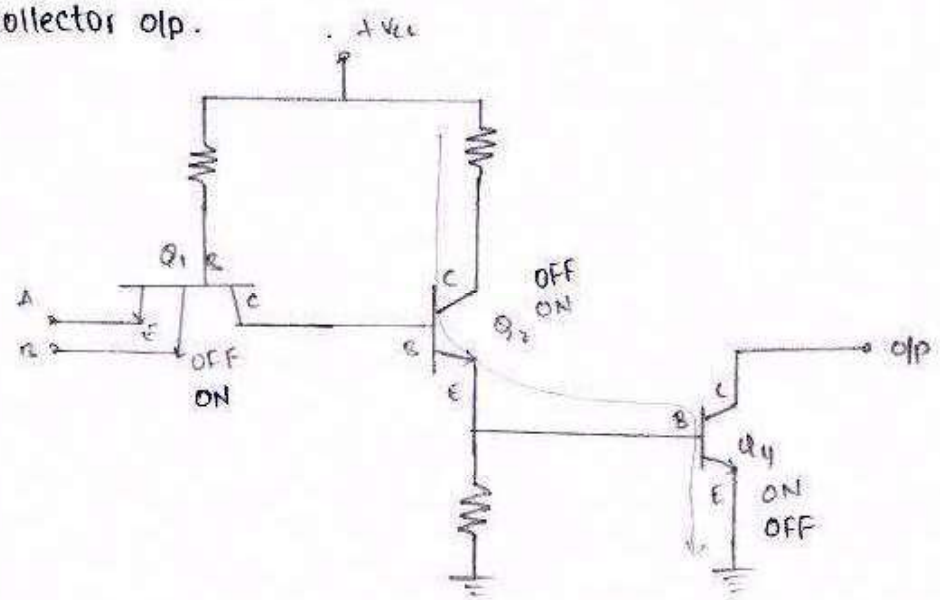


fig: open collector 2 i/p NAND gate

→ When Q_4 is ON, o/p is low. Similarly when Q_4 is OFF, there is no o/p at the collector of Q_4 transistor. So the circuit will not work properly untill you connect an external pull-up resistor.

→ It is shown in below fig.

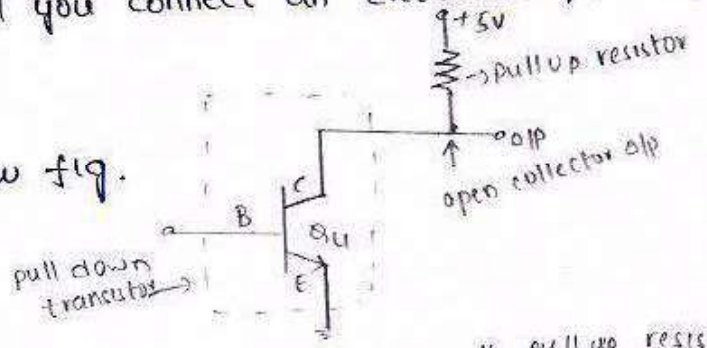


fig: Open collector with pull up resistor

→ Now when Q_4 is ON, o/p is low. when Q_4 is OFF, o/p is high because of o/p is tied to V_{cc} through an external pull-up resistor.

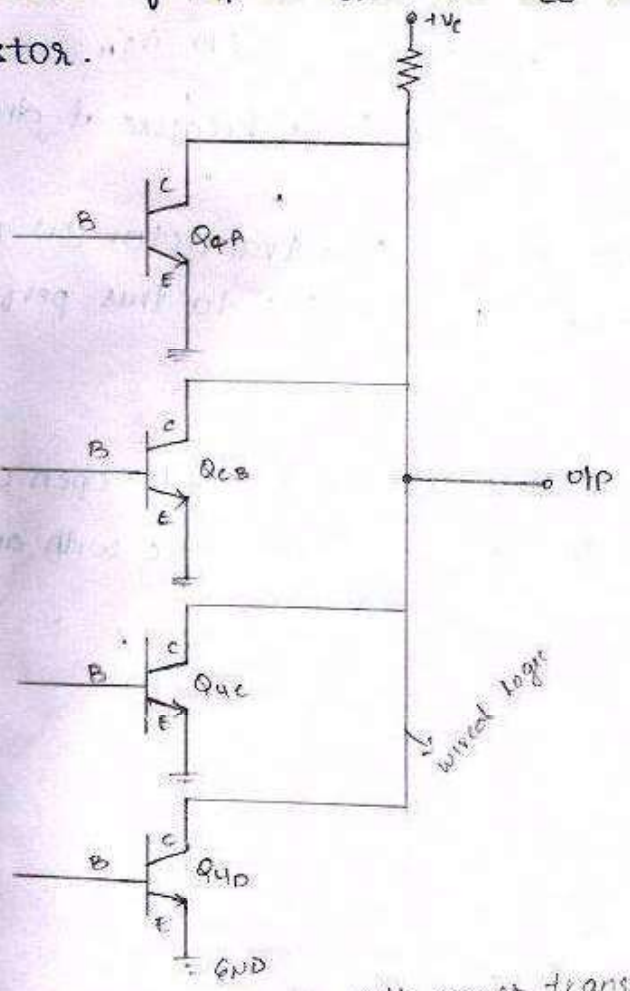
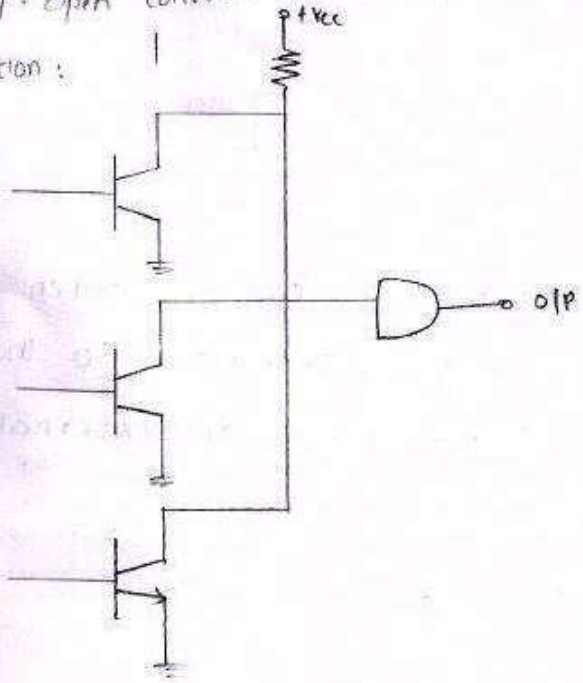


fig: open collector with no. of transistors

- Wired AND Connection:

→



→ The open collector o/p's are connected together by a AND gate is called as Wired AND connection.

The above fig. shows the wired AND connection with open collector o/p's.

It shows that the o/p is high, only when all switches are open. i.e., ∞ only when o/p of each stage is high.

Thus the o/p is the logical AND operation of the logic function performed by the gates.

Applications of Open Collector o/p's :-

1. It is used in driving lamps (or) relays.
2. It is used in performing wired logic.
3. It is used in construction of common bus system.

Tri-State TTL :

→ Tri state means that 3 o/p states i.e., low state, high state & high impedence state (Z).

→ The logic symbol for Tri state is shown in below fig.

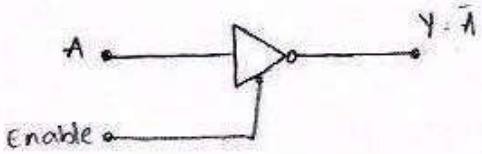
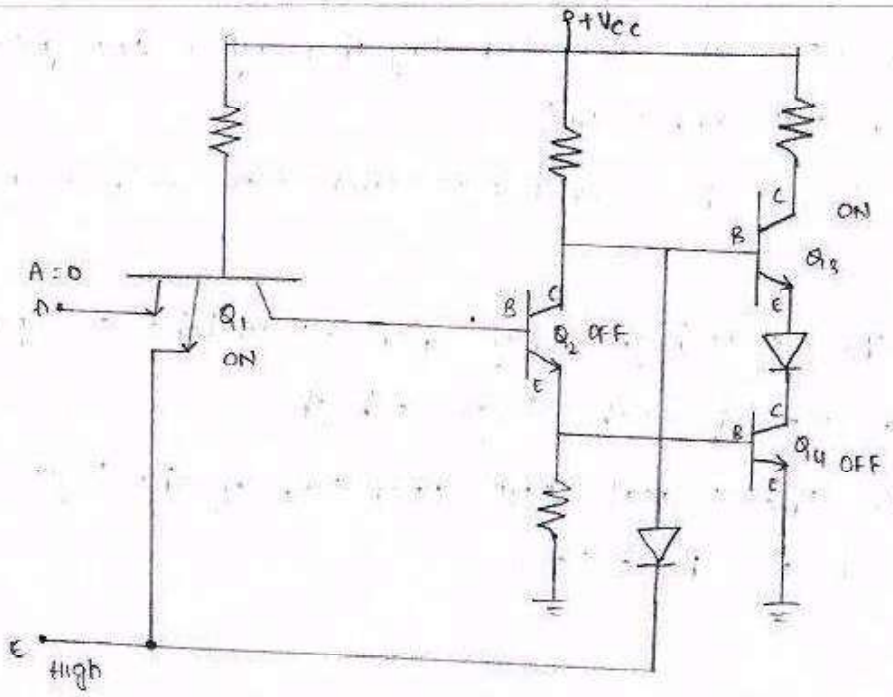


fig: Logic Symbol for Tri state when active low enable

The below figure shows the ~~transc~~ tri state TTL circuit.



→ When enable pin is high & A = 1, the transistor Q_1 is OFF and Q_2 is ON. Due to this Q_4 is going to be ON. So O/P 'Y' is low.

→ When enable pin is high & A = 0, the transistor Q_1 is ON, Q_2 is OFF, due to this Q_4 is OFF. So O/P Y is high.

→ When enable pin is low, the I/P pin A is either Low or high Q_1 is ON, Q_2 is OFF, Q_4 is OFF & Q_3 is going to OFF because the base current of Q_3 is going to ground. So here when enable pin is low the Q_3 & Q_4 transistors are in OFF state. So we get the high impedence state at the O/P.

Enable	input A	output Y
High	0	High
High	1	low
Low	0	High impedance state (Z)
Low	1	Z

MOS Logic :

→ MOS transistors is a MOSFET group. It contains 3 categories.

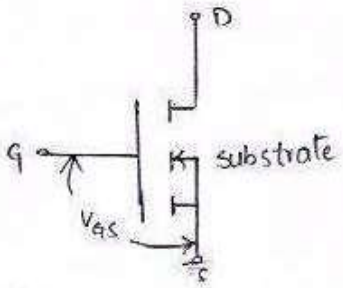
1. NMOS
2. PMOS
3. CMOS

1. NMOS :

→ NMOS uses N-channel MOSFET. In NMOS MOSFET electrons are the majority charge carriers. So in this device current carriers are electrons.

→ Electrons are moving faster than holes. So NMOS device widely is used in digital systems.

→ The below fig. shows the symbol of an N-channel MOSFET.



→ It contains 3 terminals namely gate, source, drain.

→ Gate terminal can be acts as an i/p terminal & drain terminal acts as an o/p terminal & source is grounded for drain & gate.

→ Here i/p is applied b/w gate & source. The NMOS device is ON when i/p is high (5V) & it is OFF when i/p is low (0V). i.e.,

* If $V_G = 5V$ (high)

we know that, $V_{GS} = V_G - V_S$ where $V_S = 0$ [∵ source is grounded]

$= 5 - 0$

$$\boxed{V_{GS} = 5V}$$

* if $V_G = 0V$

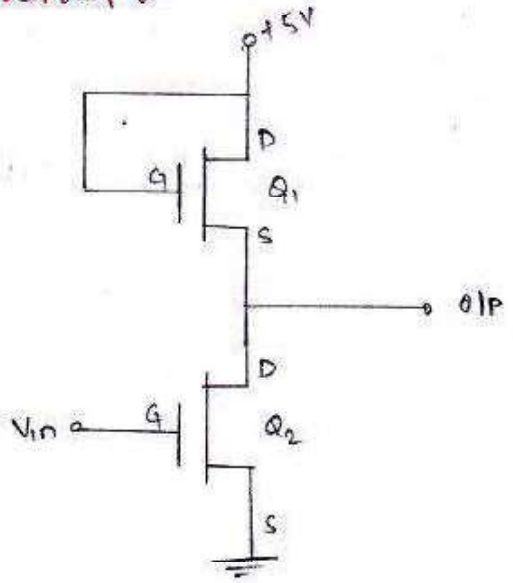
$V_{GS} = V_G - V_S$

$= 0 - 0$

$\boxed{V_{GS} = 0V}$

→ The above expression says that i/p is high device is ON, because the voltage drop b/w gate & source is 5V. The i/p is low device is OFF because the voltage drop b/w gate & source is 0V.

NMOS Inverter :



→ The above fig. shows the NMOS inverter. Gate terminal of Q_1 transistor is connected to +5V then Q_1 transistor is always ON.

→ When Q_2 transistor is switched based on i/p's.

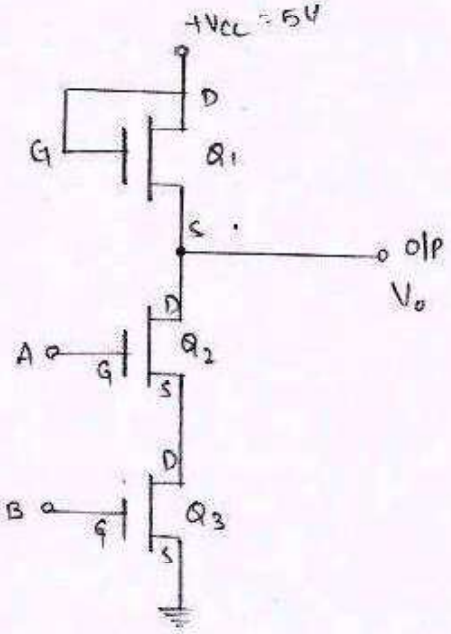
→ When $V_{in} = 0V$, Q_2 is off, so o/p is high.

→ When $V_{in} = 5V$, Q_2 is ON, so o/p is low.

V_{in}	Q_1	Q_2	o/p
Low	ON	OFF	High
High	ON	ON	Low

Two input NMOS NAND Gate :

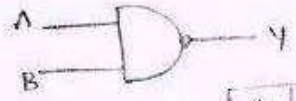
	Product	Sum
NMOS	series	parallel



Function Table :-

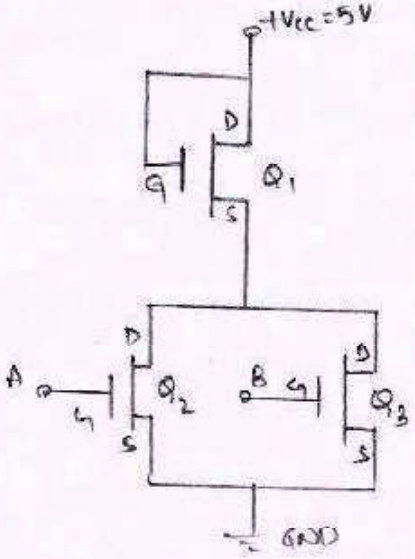
A	B	Q_1	Q_2	Q_3	V_o
0	0	ON	OFF	OFF	High
0	1	ON	OFF	ON	High
1	0	ON	ON	OFF	High
1	1	ON	ON	ON	Low

NAND Gate



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

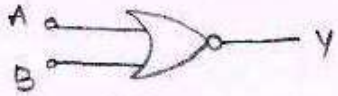
Two input NMOS NOR Gate :-



Function Table

A	B	Q_1	Q_2	Q_3	V_o
0	0	ON	OFF	OFF	High
0	1	ON	OFF	ON	Low
1	0	ON	ON	OFF	Low
1	1	ON	ON	ON	Low

NOR



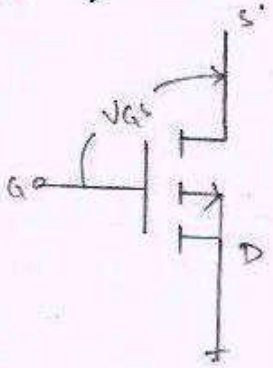
$Y = \overline{A+B}$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

2. PMOS :

→ PMOS uses P-channel MOSFET. In this device holes are the majority charge carriers. So current carriers are holes.

→ The symbol of PMOSFET shown in below figure.



→ Here gate acts as an i/p terminal, applied voltage is fed to the source terminal.

→ when i/p is low at gate terminal, device is ON.

→ when i/p is high at gate terminal, the device is OFF.

* i.e., V_G = High (5V)

We know that $V_{GS} = V_G - V_S$

$\therefore V_S = 5V$

$\Rightarrow V_{GS} = 5 - 5$

$\boxed{V_{GS} = 0V}$

* If V_G = 0V (low)

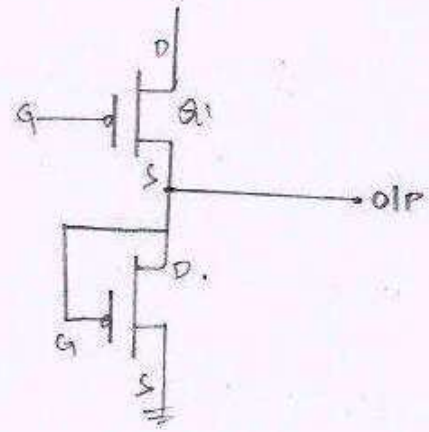
$V_{GS} = 0 - 5$

$\boxed{V_{GS} = -5V}$

→ The above expression says that -ve voltage is needed to turn ON the device.

PMOS Inverter :

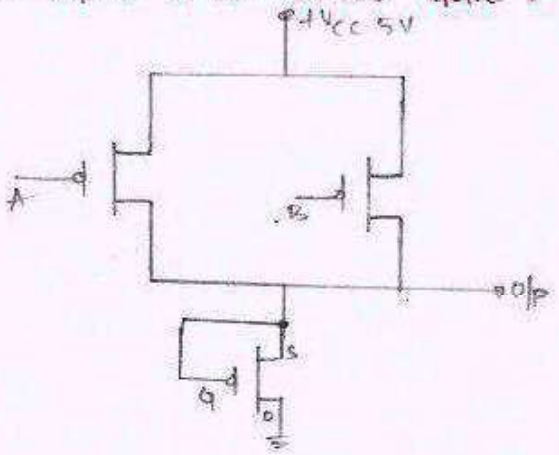
	NAND	NOR
	product	Sum
pmos	parallel	series



Function-table

V_{IN}	Q_1	Q_2	V_O
Low	ON	OFF	High
High	OFF	ON	Low

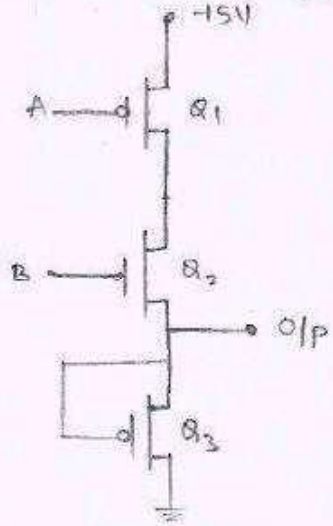
Two input PMOS NAND Gate :-



Function-table

A	B	Q_1	Q_2	Q_3	V_O
0	0	ON	ON	OFF	High
0	1	ON	OFF	OFF	High
1	0	OFF	ON	OFF	High
1	1	OFF	OFF	ON	Low

Two input PMOS NOR Gate :-



Function-table

A	B	Q_1	Q_2	Q_3	V_O
0	0	ON	ON	OFF	High
0	1	ON	OFF	ON	Low
1	0	OFF	ON	ON	Low
1	1	OFF	OFF	ON	Low

CMOS Logic :

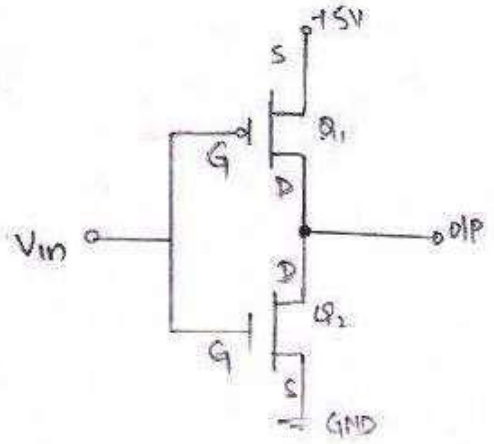
→ CMOS is a complimentary metal Oxide Semiconductor device.

→ CMOS circuit can be designed by series connection of PMOS & NMOS transistors.

→ CMOS circuit has low power dissipation. So this circuit is widely used in digital circuits.

→ CMOS can also be acts as an inverter circuit.

CMOS Inverter :-

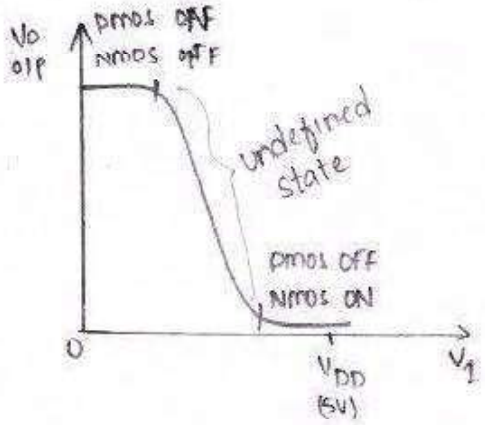


V_{in}	Q_1	Q_2	V_0
LOW	ON	OFF	High
High	OFF	ON	Low

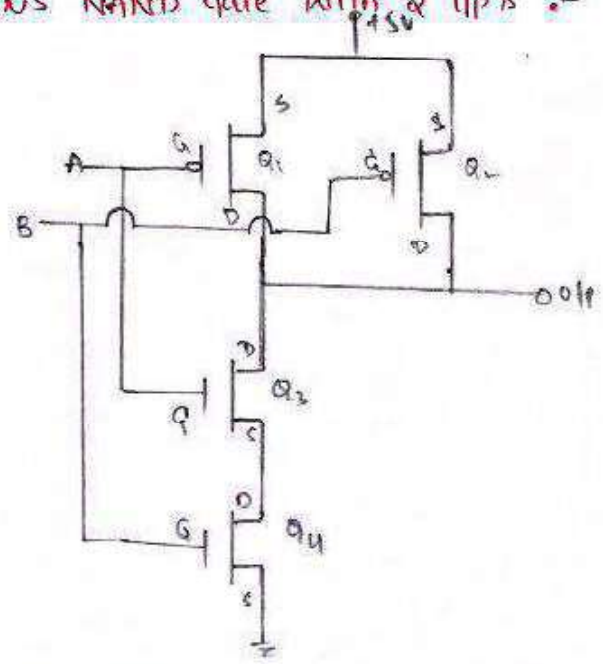
→ When V_{in} = 0V, Q_1 is ON & Q_2 is OFF. So O/P is high.

→ When V_{in} = 5V, Q_1 is OFF & Q_2 is ON. So O/P is low.

→ The below fig shows the voltage characteristics b/w I/P & O/P.



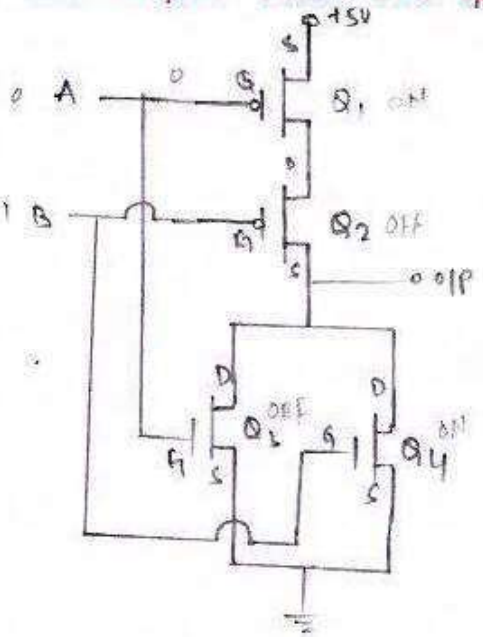
CMOS NAND Gate with 2 i/p's :-



Function table

A	B	Q_1	Q_2	Q_3	Q_4	V_o
0	0	ON	ON	OFF	OFF	High
0	1	ON	OFF	OFF	ON	High
1	0	OFF	ON	ON	OFF	High
1	1	OFF	OFF	ON	ON	LOW

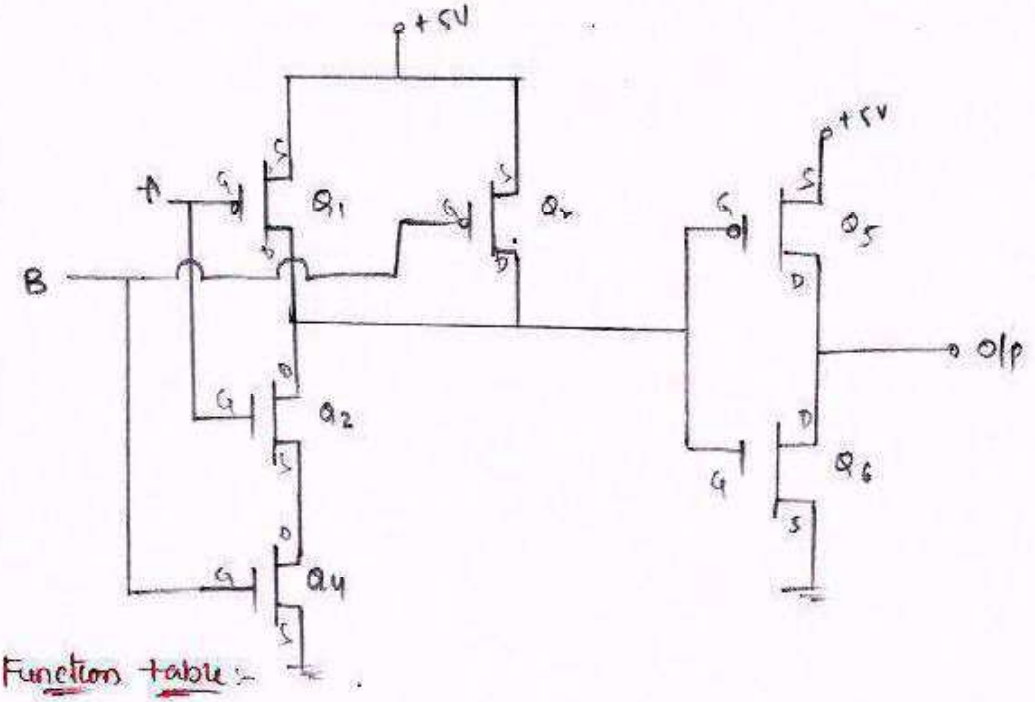
Two input CMOS NOR Gate :-



Function table

A	B	Q_1	Q_2	Q_3	Q_4	V_o
0	0	ON	ON	OFF	OFF	High
0	1	ON	OFF	OFF	ON	low
1	0	OFF	ON	ON	OFF	low
1	1	OFF	OFF	ON	ON	Low

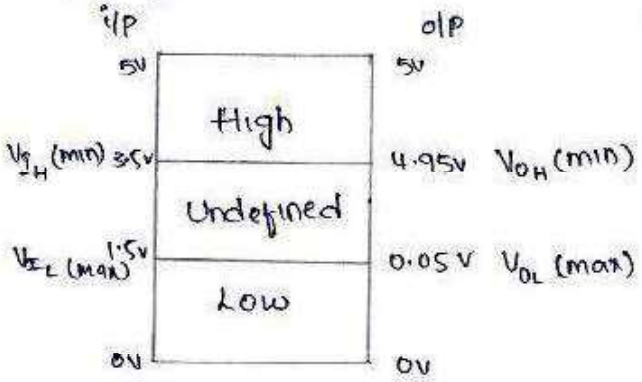
CMOS AND Gate :-



Function table :-

A	B	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	V_o
0	0	ON	ON	OFF	OFF	OFF	ON	Low
0	1	ON	OFF	OFF	ON	OFF	OFF	LOW
1	0	OFF	ON	ON	OFF	ON	ON	High
1	1	OFF	OFF	ON	ON	OFF	OFF	high

CMOS Logic gate Levels :-



Where, $V_{I_H}(min)$ - it is minimum i/p voltage in the high state is 3.5V

$V_{I_L}(max)$ - it is maximum i/p voltage in the low state is 1.5 V

$V_{O_H}(min)$ – it is minimum o/p voltage in the high state is 4.95 V

$V_{O_L}(max)$ – it is maximum o/p voltage in the low state is 0.05 V.

Noise Margin :

It is defined as the difference blw i/p & o/p voltage levels of CMOS def device. It is 2 types.

1. High state noise margin (V_{N_H})
2. Low state noise margin (V_{N_L})

1. High state Noise Margin :-

It is defined as the difference of the minimum o/p voltage in the high state & minimum i/p voltage in the high state is given by $V_{N_H} = V_{O_H}(min) - V_{I_H}(min)$

$= 4.95\ V - 3.5V$

$$\boxed{V_{N_H} = 1.45\ V}$$

Low state Noise Margin :-

It is defined as the difference of the maximum o/p voltage in the low state & maximum i/p voltage in the low

state i.e given by

$$V_{NL} = V_{IL}(max) - V_{OL}(max)$$

$$= 1.5\,V - 0.05\,V$$

$$\boxed{V_{NL} = 1.45\,V}.$$

i/p logic voltage level o/p logic level

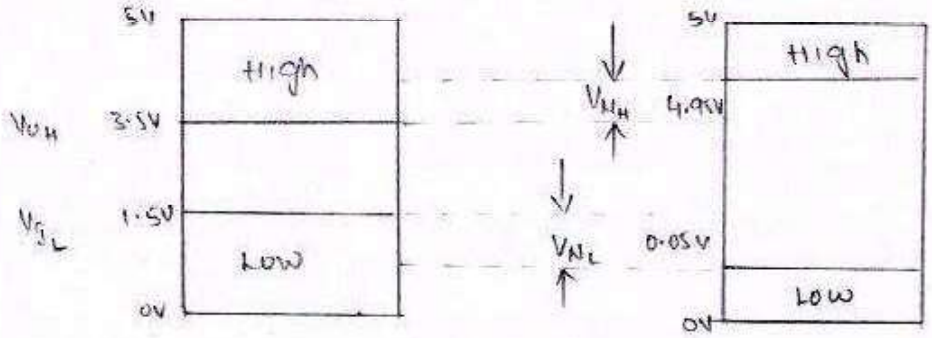


fig: CMOS Noise Margin

CMOS Open Drain Output :-

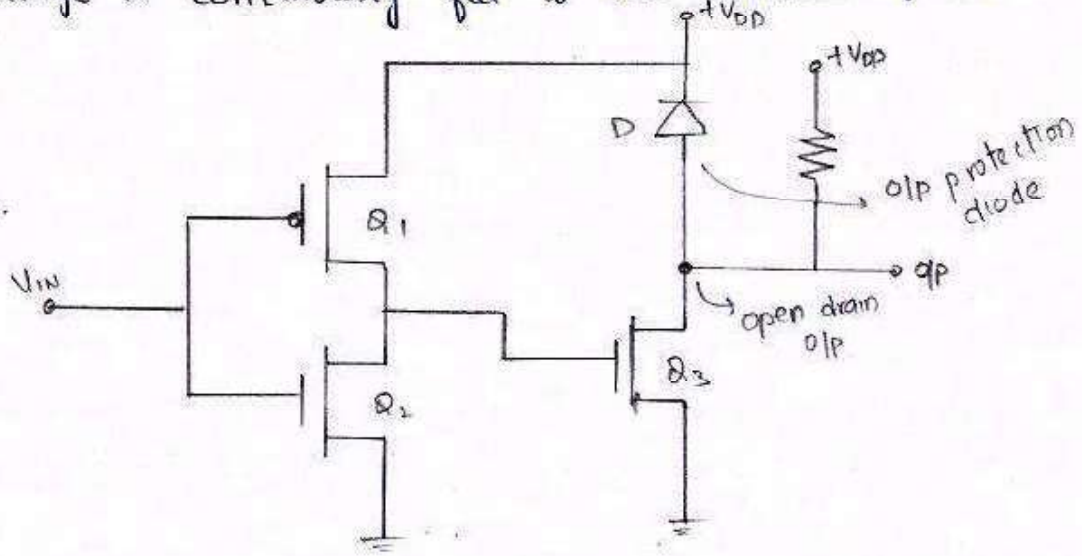
→ The below circuit shows the CMOS open drain output.

Here we have to take the o/p at the open drain of Q_3 transistor.

→ When o/p is high the resistor is called as pull-up resistor.

→ The diode 'D' is used to product the o/p.

→ The diode 'D' is always reverse bias because of applied voltage is continously fed to the cathode of diode.



→ When i/p is low Q_1 is ON & Q_2 is OFF due to this Q_3 is ON. So o/p is low.

→ When i/p is high Q_1 is OFF, Q_2 is ON due to this Q_3 is OFF & o/p is high.

CMOS Tri-state Output :-

→ Some cmos device uses control i/p that can be used to disable both the p-channel & N-channel transistors.

→ When both the transistors are disable the o/p has a 3rd state i.e., not a logic state at all, it is called high impedence (Hi-z) (or) floating state (or) Tri-state.

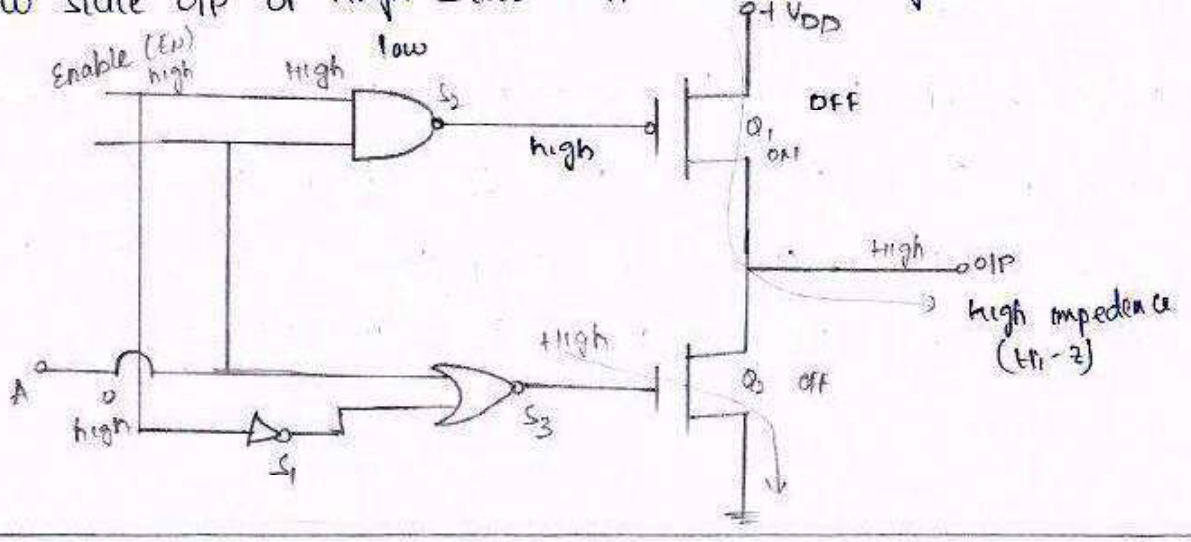
→ Thus cmos devices with enable i/p can have o/p in one of the three states logic 0, logic 1 & tristate or high impedence state.

→ Tri-state o/p's are used to create a tri-state bus.

CMOS tri-state output buffer :-

→ Here enable pin can be acts as an enable i/p that can be used to produce high

→ When enable pin is deactivated the circuit produces either low state o/p or high state o/p based on given i/p.



EN	A	S_1	S_2	S_3	Q_1	Q_2	o/p
High	0	low	high	high	OFF	ON	low
High	1	low	low	high	ON	OFF	High
Low	1	High	High	low	OFF	OFF	Hi-Z
Low	1	High	High	low	OFF	OFF	Hi-Z

→ When enable pin is high i/p A = 0, the o/p of NOT gate S_1 is low & o/p of NAND gate S_2 is high similarly o/p of NOR gate S_3 is high due to this configuations Q_1 is OFF & Q_2 is ON due to this o/p is low.

→ When enable pin is high i/p A = 1, the o/p of NOT gate S_1 is low & o/p of NAND gate S_2 is low, o/p of NOR gate S_3 is high due to this configuration Q_1 is ON & Q_2 is OFF due to this o/p is high.

→ When enable pin is low i/p A = 0, the o/p of NOT gate S_1 is high & o/p of NAND gate S_2 is high, o/p of NOR gate S_3 is low due to this configuration Q_1 is off & Q_2 is OFF. Due to this o/p is high impedence (Hi-Z)

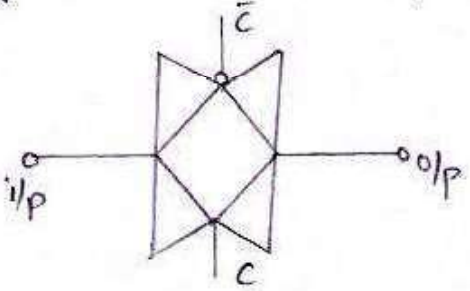
→ When enable pin is low i/p A = 1, the o/p of NOT gate S_1 is high & o/p of NAND gate S_2 is high, o/p of NOR gate S_3 is low. Due to this configuration Q_1 is OFF, Q_2 is OFF. Due to this o/p is high impedence.

TRANSMISSION GATE :

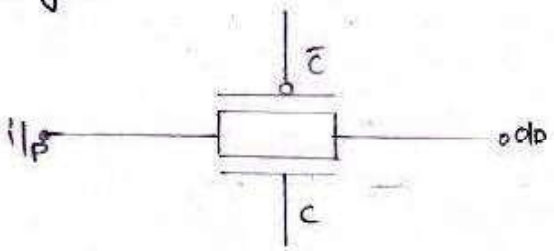
→ Transmission gate can be designed by parallel connection of PMOS & NMOS transistors. It can be acts an digital switch.

→ It is also called as complimentary CMOS switch.

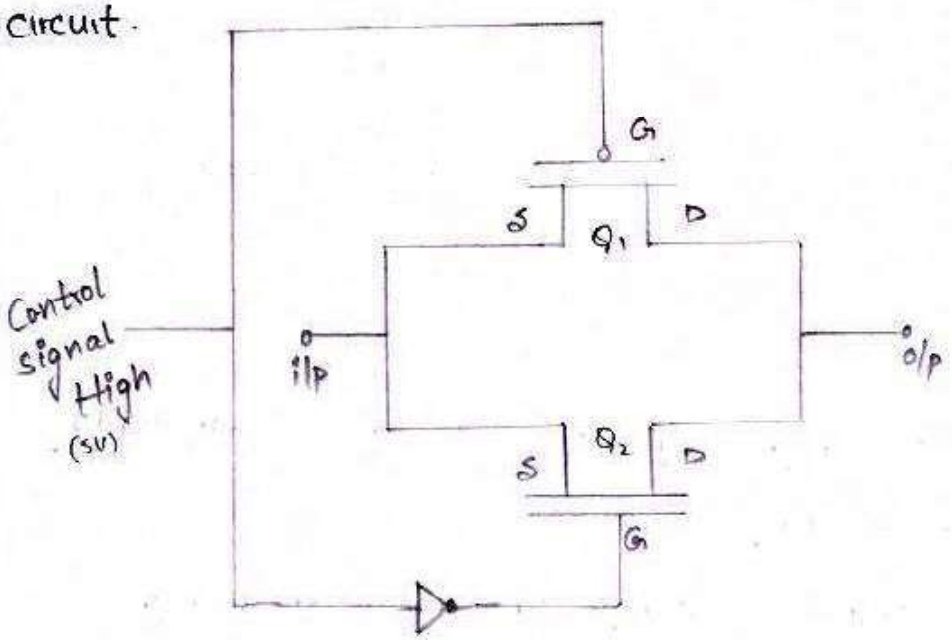
→ The below fig shows the circuit symbol of Transmission gate.



→ The below fig. shows the circuit schematic representation of transmission gate.



→ Where c & $\bar{c}$ represents the complimentary control signals of the circuit.



→ The above fig shows the operation of transmission gate when control i/p is high.

if data i/p is 0V:

i.e., $V_s = 0V$.

PMOS : $V_g = 5V$; $V_s = 0V$

W.K.T $V_{gs_1} = V_g - V_s$

$= 5 - 0$

$V_{gs_1} = 5V$

NMOS : $V_g = 0V$; $V_s = 0V$

W.K.T $V_{gs_2} = V_g - V_s$

$= 0 - 0$

$V_{gs_2} = 0V$

→ If the data i/p is low V_{gs_1} is +ve voltage & V_{gs_2} is 0V. so both transistors is going to OFF.

→ Similarly if the data i/p is 5V (or) high

PMOS : $V_g = 5$; $V_s = 5V$

W.K.T $V_{gs_1} = V_g - V_s$

$= 5 - 5$

$= 0V$

NMOS : $V_{gs_2} = V_g - V_s$ $V_g = 0V$, $V_s = 5V$

$= 0 - 5V$

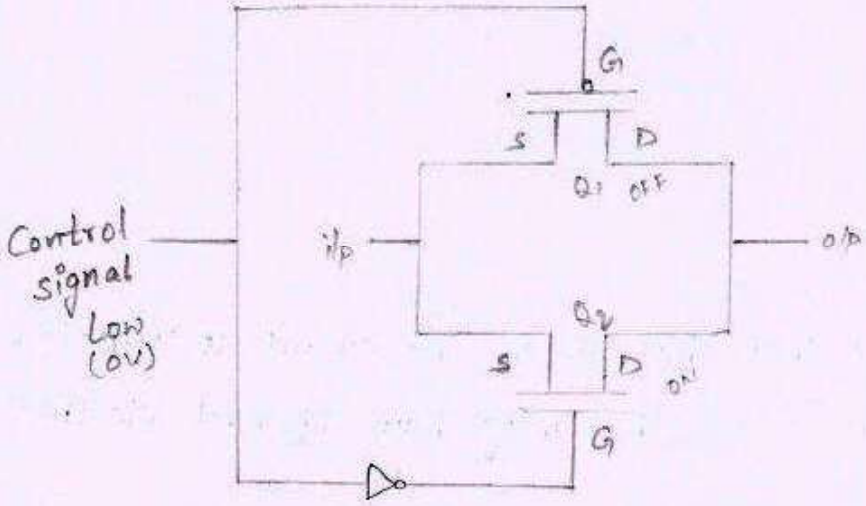
$V_{gs_2} = -5V$

→ If the data i/p is high, V_{gs_1} is 0V & V_{gs_2} is -ve voltage. so both transistors is going to be OFF.

→ Therefore when control i/p is high, the device is in high

impedence state (Hi-Z)

→ Now, suppose that control i/p is low. It is shown in below fig.



→ If the data i/p is low, V_{gs_1} is 0V, V_{gs_2} is +ve voltage (5V). Therefore Q_1 transistor is OFF & Q_2 transistor is ON.

→ So the current flowing through the Q_2 transistor.

→ similarly when data i/p is high,

PMOS : $V_{gs_1} = V_g - V_s$

$= 0 - 5$

$\boxed{V_{gs_1} = -5V}$

NMOS : $V_{gs_2} = V_g - V_s$

$= 5 - 5$

$\boxed{V_{gs_2} = 0V}$

→ When data i/p is high, V_{gs_1} is -ve, & V_{gs_2} is 0V. SO Q_1 is ON & Q_2 is OFF. SO current flows through Q_1 transistor.

→ Thus there is always conduction path from i/p to o/p when control i/p is low.

Truth table :-

Control slL	ilp	Q_1	Q_2	olp
High	0	OFF	OFF	Hi-Z
High	1	OFF	OFF	Hi-Z
Low	0	OFF	ON	High
Low	1	ON	OFF	High

IC Interfacing :

Interfacing means that the olp of one circuit is connected to the ilp of another circuit that may have different electrical characteristics.

→ When 2 circuits have different electrical characteristics direct connection cannot be made.

→ In such cases driver & load circuits are connected through interface circuit.

→ One must consider the following important points while interfacing of 2 circuits

1. The driver olp must satisfy the voltage & current requirements of the load circuit.
2. The driver & load circuit may require different power supplies.

TTL Driving CMOS :

→ Here TTL is the driver circuit and CMOS is a load circuit.

→ The 2 circuits are from different families with different electrical characteristics. Therefore we must check that the driving device can meet the current & voltage requirements of the load device.

i/p & o/p Current values		i/p & o/p voltage values	
TTL	CMOS	TTL	CMOS
$I_{IH} = 40\mu A$	$I_{IH} = 1\mu A$	$V_{IH(min)} = 2\ V$	$V_{IH(min)} = 3.5\ V$
$I_{OH} = 400\mu A$	$I_{OH} = 4mA$	$V_{OH(min)} = 2.4\ V$	$V_{OH(min)} = 4.95\ V$
$I_{IL} = 1.6mA$	$I_{IL} = 1\mu A$	$V_{IL(max)} = 0.8\ V$	$V_{IL(max)} = 1.5\ V$
$I_{OL} = 16mA$	$I_{OL} = 4mA$	$V_{OL(max)} = 0.4\ V$	$V_{OL(max)} = 0.05\ V$

→ If we compare TTL o/p currents with the i/p currents of cmos there is no any problem to interfacing.

→ If we compare o/p vtg levels of TTL with the i/p voltage levels of CMOS there is a problem occured to interfacing because TTL o/p voltage levels is less than the CMOS i/p voltage level.

→ In such situations TTL o/p must be raised to an acceptable level for CMOS.

→ This can be done by connecting pull up resistor at the o/p of TTL.

→ Here the need of pull up resistor is to raise the o/p voltage level of the TTL.

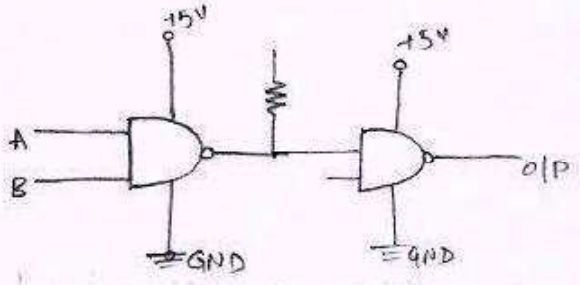


fig. TTL driving CMOS

CMOS driving TTL :-

→ Here CMOS is a driver circuit & TTL is a load circuit.

→ If we are interfacing cmos logic family with the TTL logic family we must satisfies the i/p current & i/p voltage of the TTL with the o/p voltage & o/p current of CMOS.

CMOS driving TTL in the High State :

→ Above voltage parameters shows that cmos o/p can easily supply enough voltage (V_{OH}) to satisfy the TTL i/p voltage (V_{IH}) requirement in the high state.

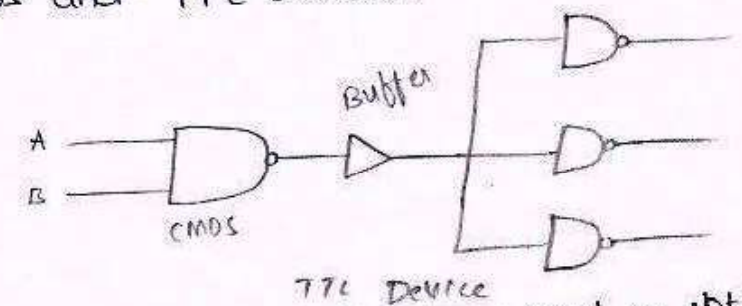
→ Similarly o/p current of CMOS is enough to satisfy the i/p current requirements of TTL. Thus no special consideration is required for the CMOS driving in the high state.

CMOS driving TTL in Low state :

→ Here CMOS o/p voltage (V_{OL}) satisfies TTL i/p voltage requirement in the low state.

→ However the current requirements in the low state are not satisfied.

→ In such situation some type of interface circuit is needed b/w the CMOS and TTL devices.



→ Here non-inverting buffer is used as interfacing circuit. It has an o/p current rating of 3mA. Which satisfies the TTL i/p current rating. requirement.

Comparision between TTL And CMOS gate Logics :-

S.No	Parameter	TTL	CMOS
1.	Device used	Transistors	PMOS & NMOS
2.	V_{OH} (min)	2.4 V	4.95 V
3.	V_{OL} (max)	0.4 V	0.05 V
4.	I_{IH} (	40μA	1μA
5.	I_{OH}	400 μA	4mA
6.	V_{IL}(max	0.8 V	1.5 V
	V_{IH}(min)	2 V	3.5 V
	I_{IL}	1.6mA	1 μA
	I_{OL}	16 mA	4 mA
	V_{NH}	1.45 V	
	V_{NL}	1.45 V	
	fan-out.	10	50
Applications :-		Digital (laboratory) instruments.	portable instruments

UNIT - V

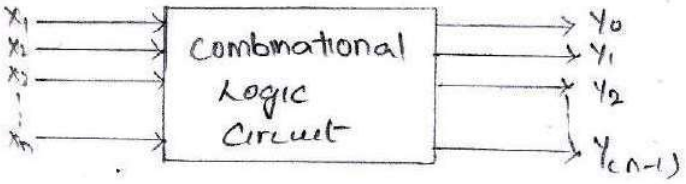
COMBINATIONAL & SEQUENCIAL CIRCUITS

Introduction :

Combinational circuit is a logic circuit in which o/p is depends on the present i/p only.

→ It does not have feedback, memory, clock signal to produce o/p signal.

→ The below fig. shows the block diagram of combinational circuit.



→ Here combinational circuit accepts 'n' i/p binary variables & generates o/p variables depending on the logical combination of gate.

→ Let 'x' be the set of 'n' i/p variables & 'y' be the 'm' o/p variables.

→ The combinational function f operate on the 'x' i/p variables to produce the 'y' o/p variables.

→ Thus the o/p function can be written as

$$Y = F(X)$$

→ Combinational circuits are used to design the decoders, encoders, multiplexers, demultiplexers etc..

Code Converters :

Code converters are required to convert one code to another code.

→ Some of these are binary to BCD, binary to Excess-3, binary to gray code and so on.

→ We have to follow the 3 steps to design the logical circuit.

Step 1: Write the truth table showing the relationship b/w i/p code & o/p code.

Step 2: For each o/p code bit determine the simplified Boolean expression using K-map.

Step 3: Realize the code converters using logic gates.

Example:- Design a 4 bit binary to BCD.

→ BCD is the number system which has the binary numbers to represent a decimal number.

→ BCD contains 10 digits (0-9).

→ In case of BCD the binary number formed by four binary digits, will be the equivalent code for the given decimal digits.

→ In BCD we can use the binary number from 0000 to 1000 only. which are the decimal equivalent from 0 to 9 respectively.

→ Suppose if we have 2 decimal digits then its equivalent BCD will be the respective 8 binary of the given decimal number. First four bits for the 1st decimal digit & next 4 bits for the next decimal digit.

Decimal	Binary				BCD				
	B_3	B_2	B_1	B_0	D_4	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	0	1
2	0	0	1	0	0	0	0	1	0
3	0	0	1	1	0	0	0	1	1
4	0	1	0	0	0	0	1	0	0
5	0	1	0	1	0	0	1	0	1

Decimal	Binary				BCD				
	B_3	B_2	B_1	B_0	D_4	D_3	D_2	D_1	D_0
6	0	1	1	0	0	0	1	1	0
7	0	1	1	1	0	0	1	1	1
8	1	0	0	0	0	1	0	0	0
9	1	0	0	1	0	1	0	0	1
10	1	0	1	0	1	0	0	0	0
11	1	0	1	1	1	0	0	0	1
12	1	1	0	0	1	0	0	1	0
13	1	1	0	1	1	0	0	1	1
14	1	1	1	0	1	0	1	0	0
15	1	1	1	1	1	0	1	0	1

Step 2: Using K-map for determining the o/p code boolean expression

For D_0

B_3B_2 \ B_1B_0	00	01	11	10
$\bar{B}_3\bar{B}_2$ 00	0	1	1	0
$\bar{B}_3B_2$ 01	0	1	1	0
B_3B_2 11	0	1	1	0
$B_3\bar{B}_2$ 10	0	1	1	0

$\therefore D_0 = B_0$

For D_1

B_3B_2 \ B_1B_0	00	01	11	10
00			1	1
01			1	1
11	1	1		
10				

$\therefore D_1 = \bar{B}_1 B_2 B_3 + B_1 \bar{B}_2 B_3$

For D_2

B_2B_2 \ B_1B_0	00	01	11	10
00				
01	1	1	1	1
11			1	1
10				

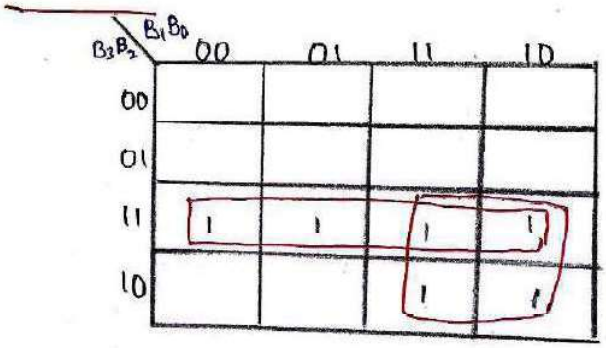
$\therefore D_2 = B_2\bar{B}_3 + B_2B_1$

For D_3

B_3B_2 \ B_1B_0	00	01	11	10
$\bar{B}_3\bar{B}_2$ 00				
$\bar{B}_3B_2$ 01				
B_3B_2 11				
$B_3\bar{B}_2$ 10	1	1		

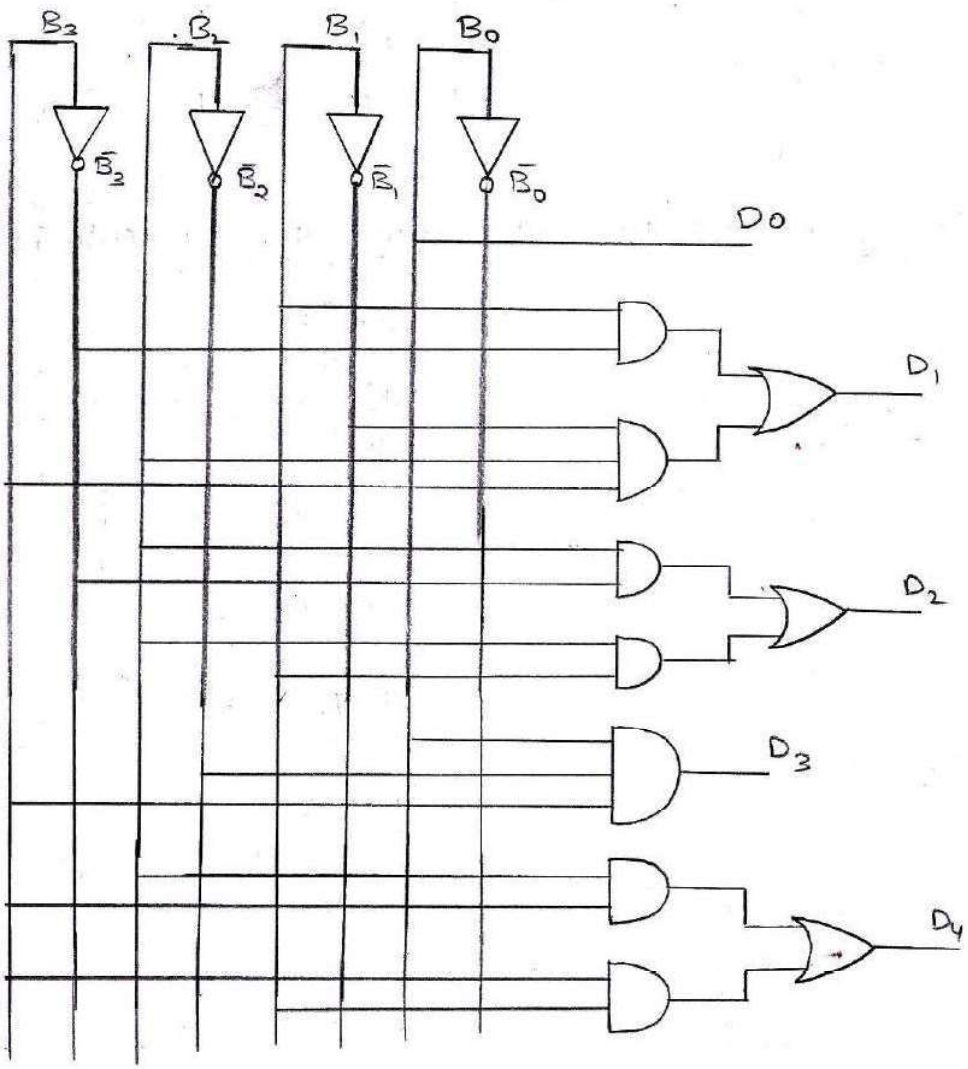
$\therefore D_3 = B_0\bar{B}_2B_3$

For D_4



$$\therefore D_4 = B_2B_3 + B_3B_1$$

Logic Diagram :-



BCD to Excess-3 Code :-

Truth table :

Decimal	BCD				Excess-3			
	B_3	B_2	B_1	B_0	E_3	E_2	E_1	E_0
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

K-map :-

For E_0

B_3B_2 \ B_1B_0	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	x	x	x	x
10	1	0	x	x

$\therefore E_0 = \bar{B_0}$

For E_1

B_3B_2 \ B_1B_0	00	01	11	10
00	1	0	1	0
01	1	0	1	0
11	x	x	x	x
10	1	0	x	x

$\therefore E_1 = \bar{B_1}\bar{B_0} + B_1B_0$

For E_2

B_3B_2 \ B_1B_0	00	01	11	10
00	0	1	1	1
01	1	0	0	0
11	X	X	X	X
10	0	1	X	X

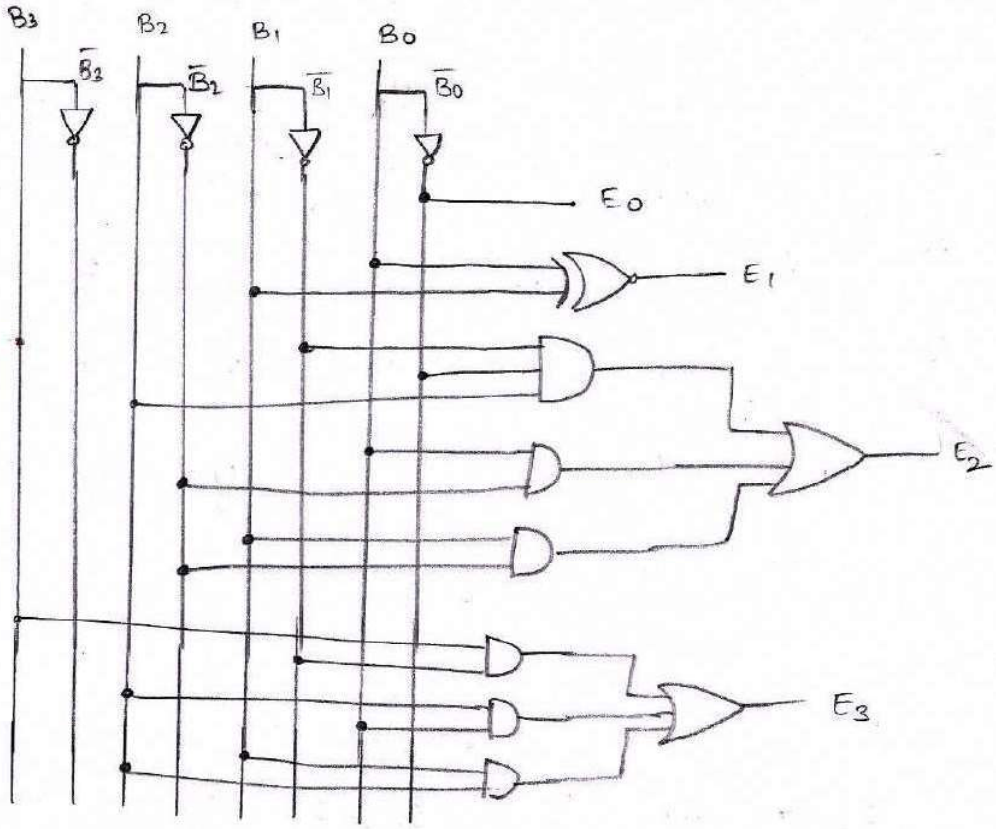
$\therefore E_2 = B_2\bar{B}_1\bar{B}_0 + \bar{B}_2B_0 + B_1\bar{B}_2$

For E_3

B_2B_2 \ B_1B_0	00	01	11	10
00	0	0	0	0
01	0	1	1	1
11	X	X	X	X
10	1	1	X	X

$\therefore E_3 = B_3\bar{B}_1 + B_2B_0 + B_1B_2$

Logic Circuit :-



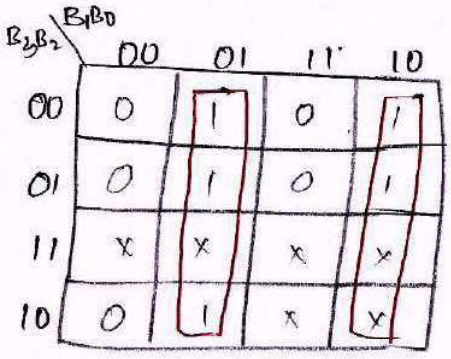
BCD to Gray Code :-

Decimal	Binary				Gray			
	B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	0	0	1
2	0	0	1	0	0	0	1	1
3	0	0	1	1	0	0	1	0

	B_3	B_2	B_1	B_0	G_3	G_2	G_1	G_0
4	0	1	0	0	0	1	1	0
5	0	1	0	1	0	1	1	1
6	0	1	1	0	0	1	0	1
7	0	1	1	1	0	1	0	0
8	1	0	0	0	1	1	0	0
9	1	0	0	1	1	1	0	1

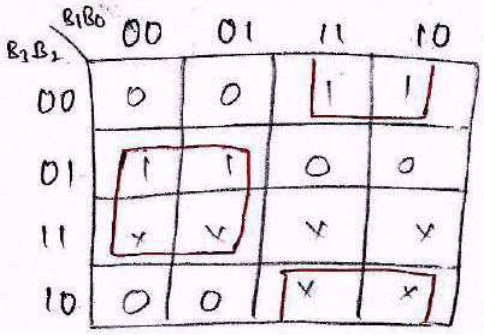
K-map :

For G_{10}



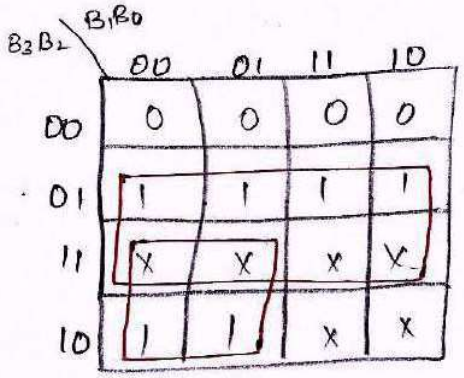
$\therefore G_0 = \bar{B_1} B_0 + B_1 \bar{B_0}$

For G_1



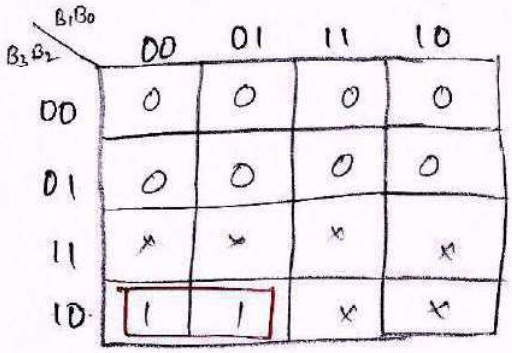
$\therefore G_1 = B_2 \bar{B_1} + B_1 \bar{B_2}$

For G_2

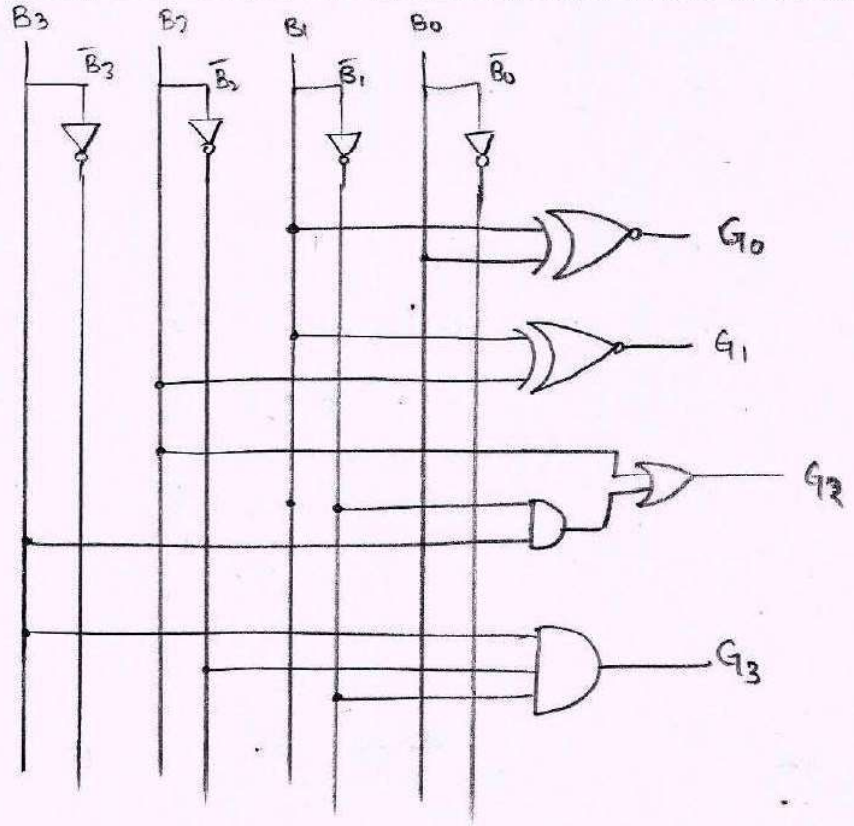


$\therefore G_2 = \bar{B_0} B_2 + B_3 \bar{B_1}$

For G_3



$G_3 = \bar{B_1} B_3 \bar{B_2}$


B3
B2
B1
B0
$\bar{B}_3$
$\bar{B}_2$
$\bar{B}_1$
$\bar{B}_0$
G_0
G_1
G_2
G_3

Decoder :-

Decoder is a combinational logic circuit, in which o/p is decoded by the i/p's.

→ A decoder is a multiple i/p, multiple o/p logic circuit that converts coded i/p into coded o/p.

Where the i/p & o/p codes are different.

→ The i/p code generally has less number of bits than the o/p bits, & there is a one-to-one mapping from i/p code to o/p code.

→ Decoder requires 'n' i/p lines & 2^n o/p lines.

Binary Decoder :-

A decoder which takes 'n' bit binary i/p's & produces 2^n o/p's from which only one o/p will be activated at a time based on the i/p lines is known as "Binary Decoder."

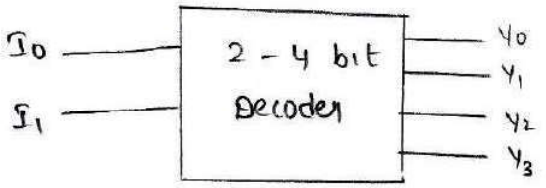
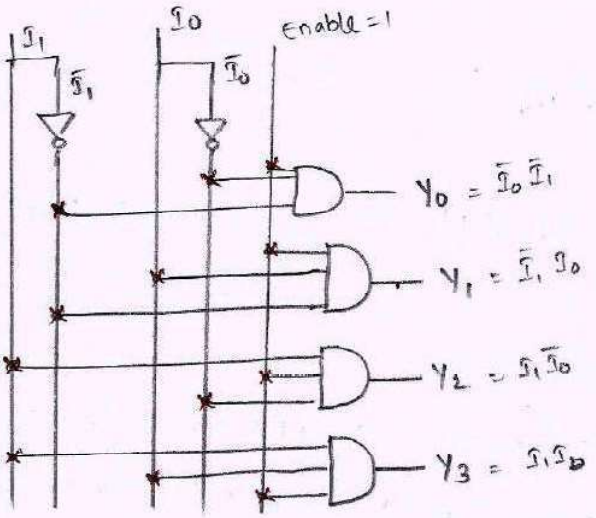


fig: Logic Symbol of 2-4 decoder

Truth table :

Enable	I_0	I_1	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

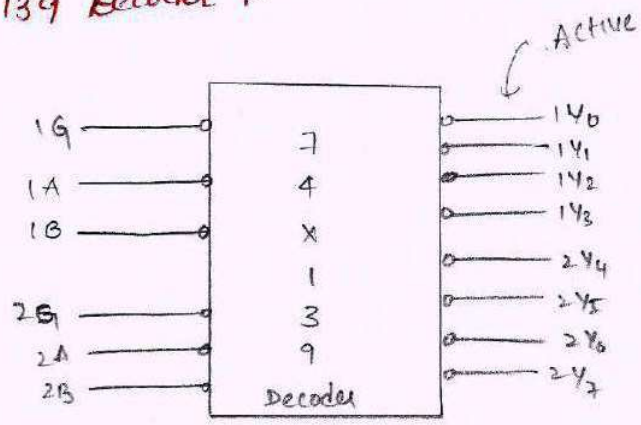


→ The i/p code word I_0, I_1 represents & Y_0, Y_1, Y_2, Y_3 are o/p code words.

→ When enable is activated then only the o/p's are decoded by i/p's. If enable is '0', all o/p's are zero.

↔

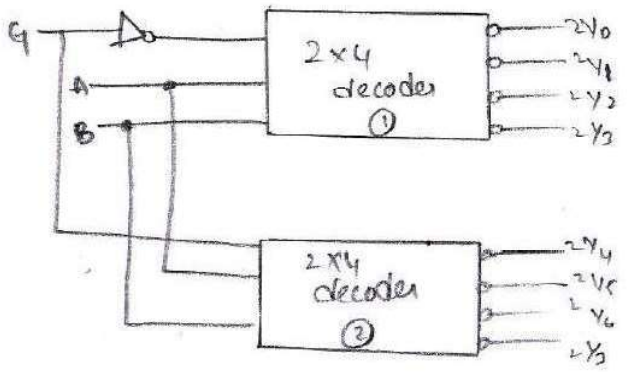
74 x 139 Decoder :-



Logic Symbol for 2 2-4 decoders

→ IC 74X139 has active low o/p's. The pin 'G' acts as an active low enable i/p.

→ Using two 2 to 4 decoders to implement 3x8 decoder circuit shown in below fig.



→ Where 'G' represents the enable pin. Then it requires active low o/p's ($1Y_0, 1Y_1, 1Y_2, 1Y_3, 2Y_4, 2Y_5, 2Y_6, 2Y_7$).

→ A & B are the i/p lines for the 2x4 decoder circuit.

→ When enable G = 0, first decoder o/p's are decoded by the i/p's.

→ When enable G = 1, second decoder o/p's are decoded by the i/p's.

→ The corresponding truth table shown in below figure.

Enable (G)	B	A	$2Y_7$	$2Y_6$	$2Y_5$	$2Y_4$	$1Y_3$	$1Y_2$	$1Y_1$	$1Y_0$
X	0	0	1	1	1	1	1	1	1	1
0	0	0	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	0	1
0	1	0	1	1	1	1	1	0	1	1
0	1	1	1	1	1	1	0	1	1	1
1	0	0	0	1	1	0	1	1	1	1
1	0	1	1	1	0	1	1	1	1	1
1	1	0	1	0	1	1	1	1	1	1
1	1	1	0	1	1	1	1	1	1	1

3x8 Decoder :-

→ 3x8 decoder is a combinational circuit. It requires 3 i/p's & 8 o/p's & one enable i/p's.

→ The logic symbol of 74x138 decoder is shown below.

→ In logic symbol A, B, C represents the i/p's & 1G, 2G, 3G represents the enable i/p's but 1G is an active high enable & 2G, 3G are active low enable i/p's.

→ $Y_0 - Y_7$ represents the active low o/p's.

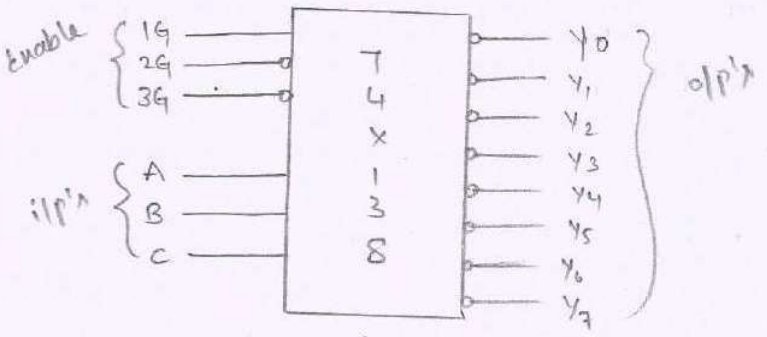
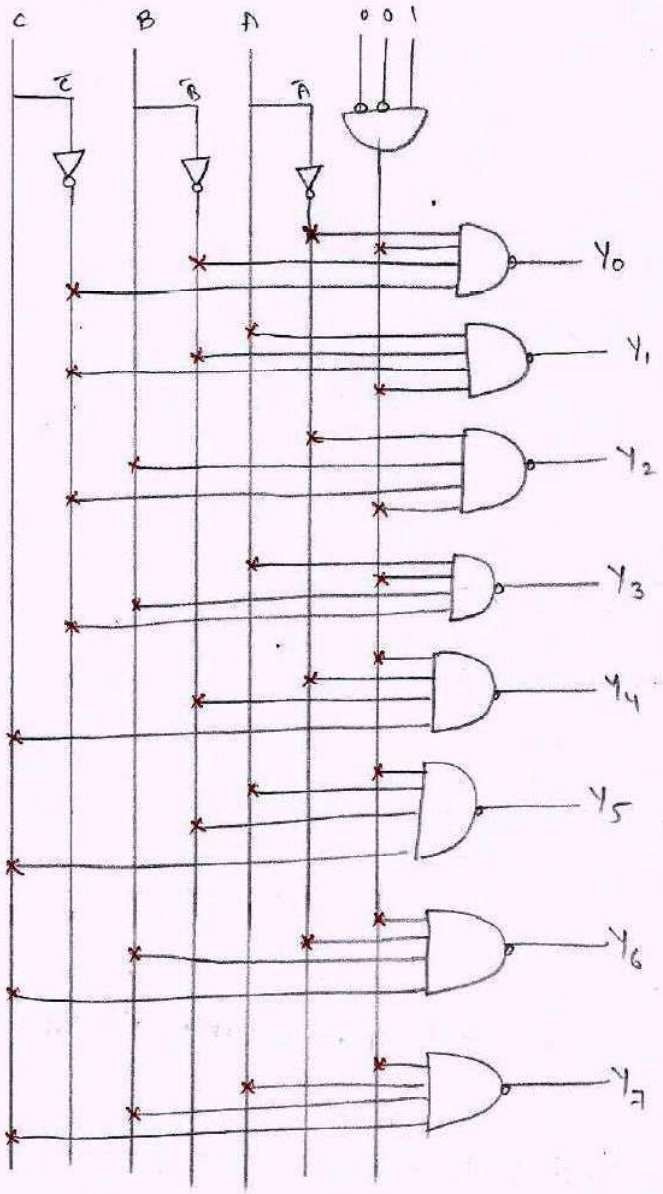


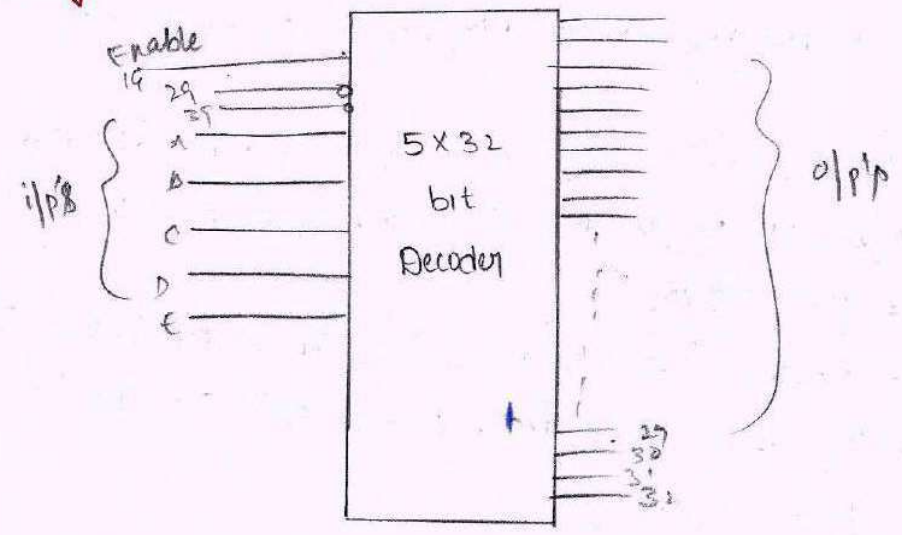
fig: Logic Symbol

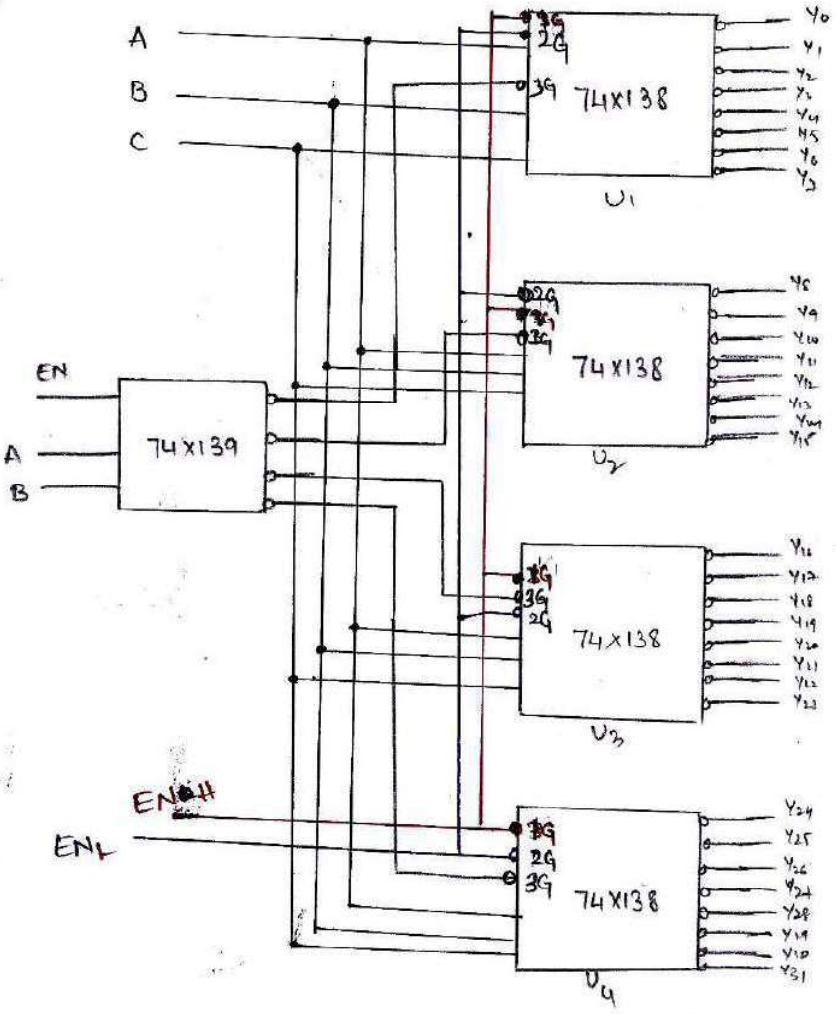
Enable			i/p's			o/p's							
1G	2G	3G	C	B	A	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
X	X	X	X	X	X	1	1	1	1	1	1	1	1
0	1	1	X	X	X	1	1	1	1	1	1	1	1
1	1	1	X	X	X	1	1	1	1	1	1	1	1
1	0	0	0	0	0	1	1	1	1	1	1	1	0
1	0	0	0	0	1	1	1	1	1	1	1	0	1
1	0	0	0	1	0	1	1	1	1	1	0	1	1
1	0	0	0	1	1	1	1	1	1	0	1	1	1
1	0	0	1	0	0	1	1	1	0	1	1	1	1
1	0	0	1	0	1	1	1	0	1	1	1	1	1
1	0	0	1	1	0	1	0	1	1	1	1	1	1
1	0	0	1	1	1	0	1	1	1	1	1	1	1

Logic Circuit :-



Design a 5 x 32 bit decoder using 4 - 74x138 IC's & 74-139 IC.





→ Here as we have 32 active low o/p's that means all o/p's are high.

→ When enable $EN_H = 1$, $EN_L = 0$ due to that 1G & 2G is activated for all decoder CKt

→ But 3G enable is not activated. It depends on 74X139 o/p pins.

→ When $EN = 0$, $N_1 = N_2 = 0$ during this i/p Y_0 should be decoded. Due to this 3G enable pin of U_1 is activated. So now o/p's $Y_0 - Y_7$ is decoded by the given i/p's.

→ Similarly when $N_1 = 0, N_2 = 1$, $EN = 0$ Y_1 should be decoded. Due to this 3G enable pin of U_2 is activated. So now o/p's $Y_8 - Y_{15}$ is decoded by the given i/p's.

→ When $N_1 = 1$, $N_2 = 0$, $EN = 0$; Y_2 should be decoded. Due to this 3G enable pin of U_3 is activated. So now o/p's $Y_{16} - Y_{23}$ is decoded by the given i/p's.

→ When $N_1 = 1$, $N_2 = 1$, $EN = 0$; Y_3 should be decoded. Due to this 3G enable pin of U_4 is activated. So now o/p's $Y_{24} - Y_{31}$ is decoded by the given i/p's.

Encoder :

Encoder is an combinational circuit. An Encoder is a digital circuit that performes the reverse operation of a Decoder.

→ An encoder has 2^n i/p's lines & 'n' o/p lines.

→ In encoder the o/p lines generate the binary code corresponding to i/p value.

→ The decoded information is presented at 2^n i/p's producing 'n' possible o/p's.

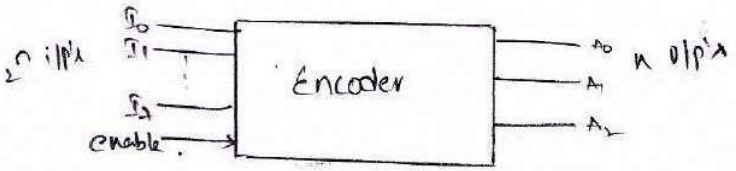


fig: general structure of an encoder

→ Let us consider 8 to 3 encoder circuit. In that 8 i/p lines & 3 o/p lines.

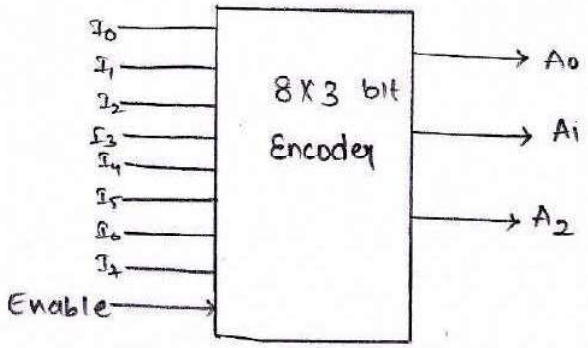


fig: 8x3 Encoder circuit

→ Here 8 i/p lines are I_0 to I_7 and o/p lines are A_0 to A_2.

→ It requires one more i/p terminal called as enable, it is used to control the entire operation of the encoder circuit. That means when enable pin is high o/p generates the binary data corresponding to the i/p data.

→ The truth table of an 8x3 bit encoder circuit shown in below fig.

Enable	I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	A_2	A_1	A_0
0	x	x	x	x	x	x	x	x	0	0	0
1	0	0	0	0	0	0	0	1	0	0	0
1	0	0	0	0	0	0	1	0	0	0	1
1	0	0	0	0	0	1	0	0	0	1	0
1	0	0	0	0	1	0	0	0	0	1	1
1	0	0	0	1	0	0	0	0	1	0	0
1	0	0	1	0	0	0	0	0	1	0	1
1	0	1	0	0	0	0	0	0	1	1	0
1	1	0	0	0	0	0	0	0	1	1	1

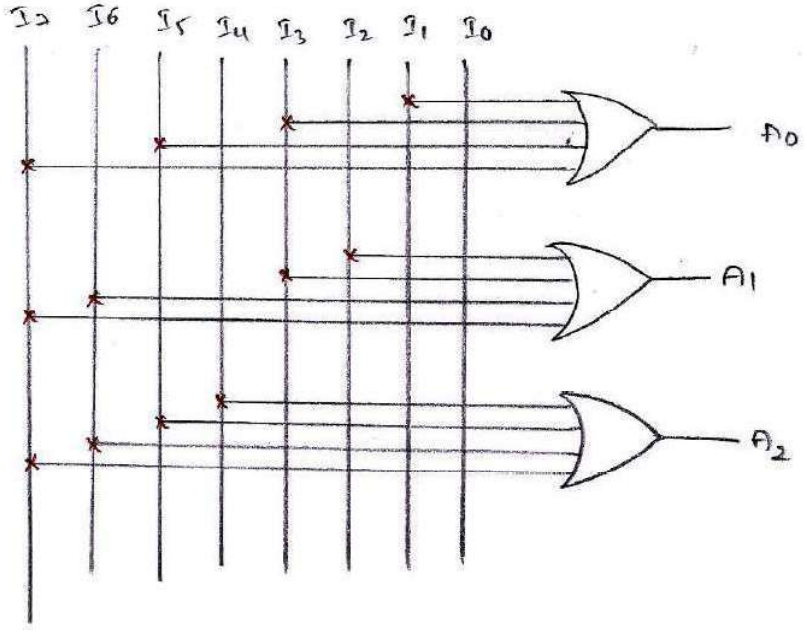
→ For an 8x3 encoder with i/p's $I_0 - I_7$ & o/p's are $A_0 - A_2$.

→ the equations can be write for the o/p's can be shown from the truth table.

$$A_0 = I_1 + I_3 + I_5 + I_7$$

$$A_1 = I_2 + I_3 + I_6 + I_7$$

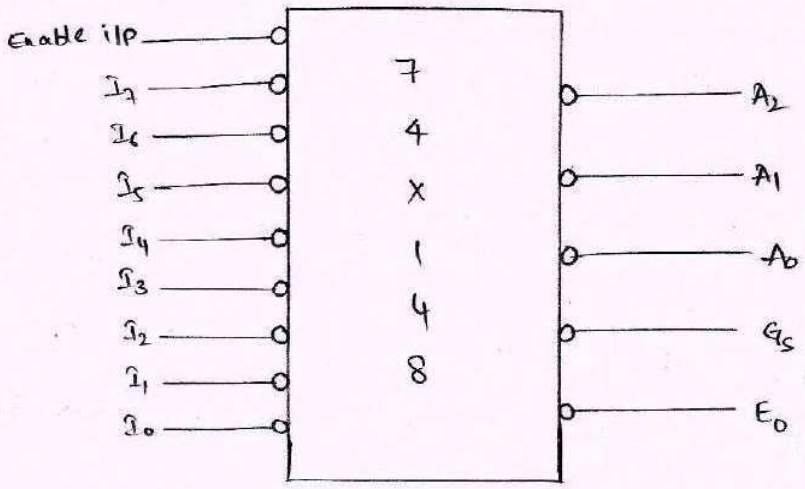
$$A_2 = I_4 + I_5 + I_6 + I_7$$



Priority Encoder :-

Priority encoder is an encoder that includes the priority functions.

Let us consider 8 i/p priority encoder, it is shown in below fig.



→ The 74x148 IC is used as an priority encoder.

→ The IC 74x148 accepts data from 8 active low i/p's & provides a BCD data on the 3 active low o/p's.

→ A priority is assigned to each i/p, so that when 2 or more i/p's are simultaneously active the o/p with the highest priority is represented on the O/P.

→ The i/p I_0 has the lowest priority & i/p I_7 has the highest priority.

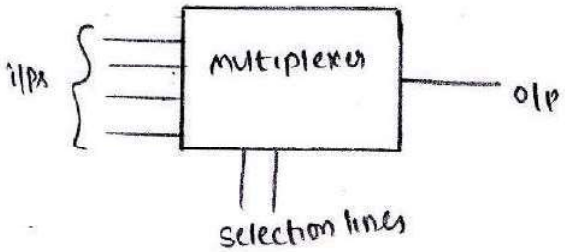
Multiplexer :-

Multiplexer is a device that allows one of several i/p signals which are to be slected and it transmits the o/p i/p which is selected in to a single line (or) o/p

→ A multiplexer has a many i/p's & one o/p.

→ A multiplexer acts as an switch.

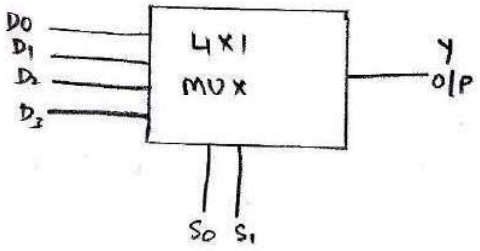
→ Multiplexer is abbreviated as "MUX". It is also known as Data selector.



→ The basic mux has several i/p lines and single o/p line. MUX consists of selection lines which are used to select the data from the i/p lines.

→ The selection of particular i/p line is controlled by a set off & selection lines.

4X1 MUX :-



Truth table:

S_0	S_1	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

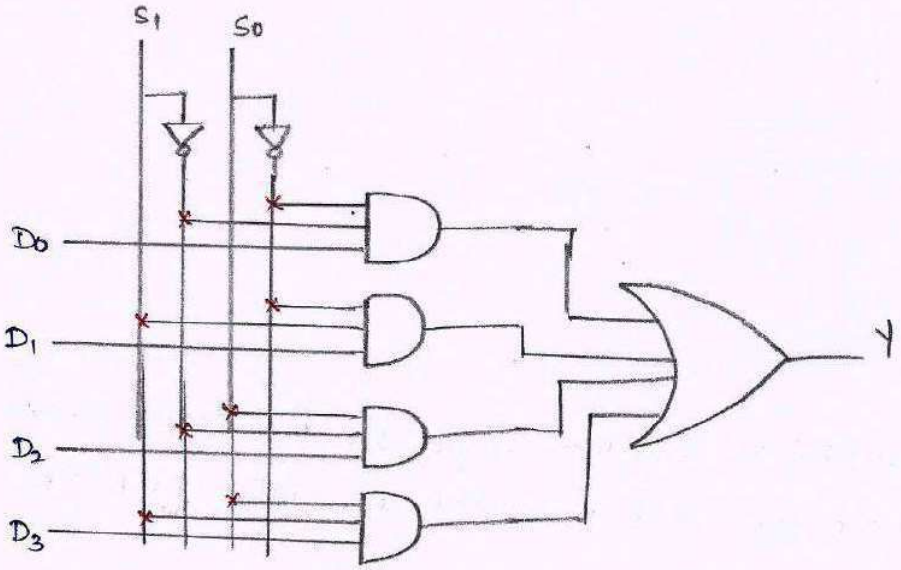
→ 4X1 MUX has 4 data i/p's i.e, D_0, D_1, D_2, D_3 and it has single o/p Y.

→ 4X1 mux consists of 2 selection lines namely S_0, S_1.

→ The selection lines are used to select the data from i/p and it transmits to the o/p.

→ For example $S_0 = 0$; $S_1 = 0$; the D_0 data i/p line can transmitted to the o/p line.

→ The corresponding logic diagram of 4X1 mux shown in below fig.



Applications :

1. Multiplexer can be used in different applications where multiple data can be transmitted using a single line.
   (a) Communication System
   (b) Telephone Network

(a). Communication Systems

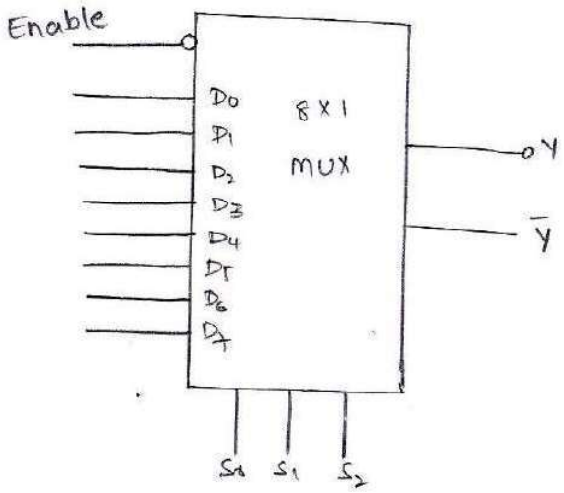
→ A MUX is used in communication system which has a transmission system.

→ A MUX is used to increase the efficiency of the communication system. by allowing the transmission of data from different channels.

2. Telephone Network :

A mux is used in telephone network to integrate the multiple audio signals on a single line of transmission.

74 x 151 IC (or) 8x1 MUX :



→ 74 x 151 is a 8x1 multiplexer. It has 8 i/p's namely D_0 to D_7.

→ It provides 2 o/p's, one is active high o/p & other is active low o/p.

→ The above fig. shows the logic symbol for 74x151.

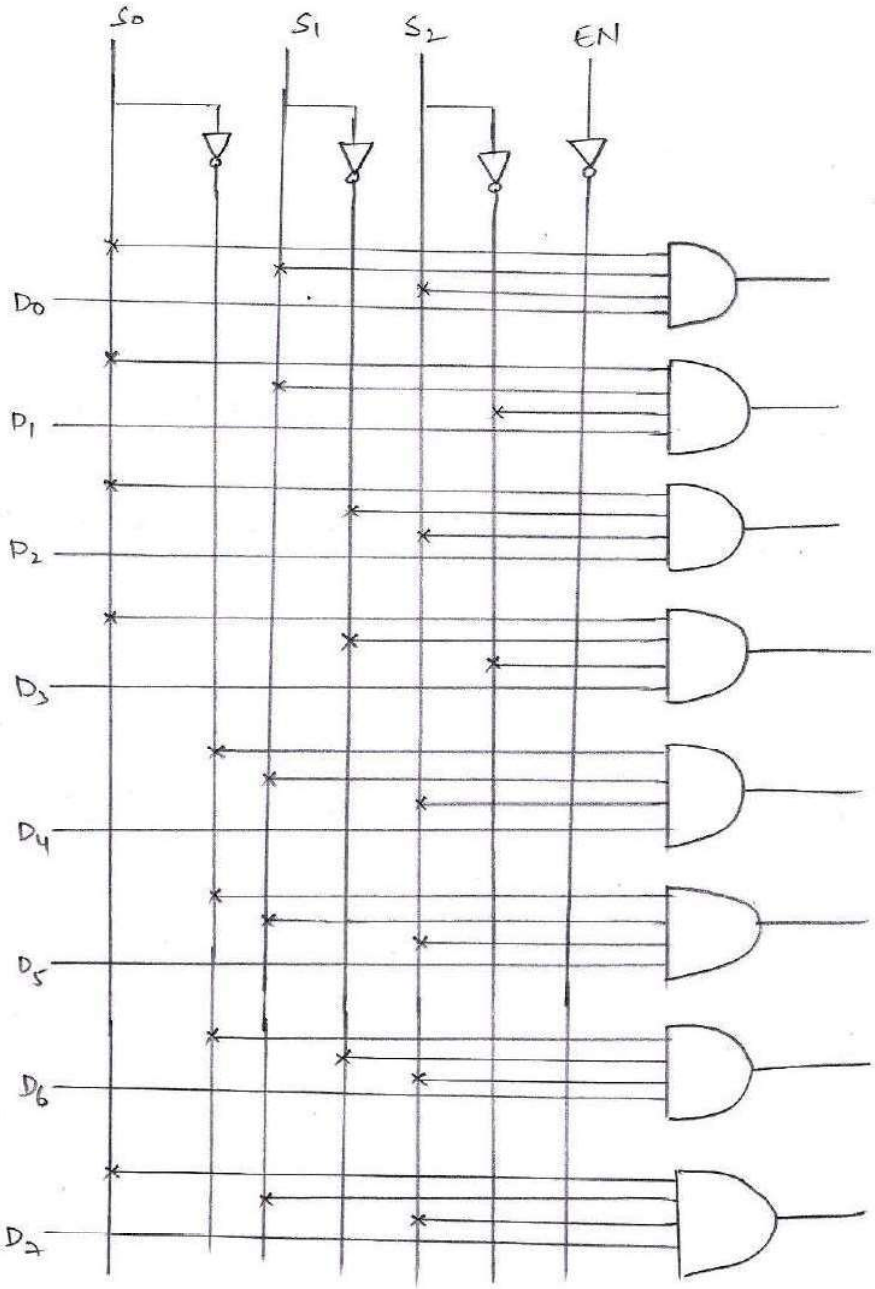
→ As shown in the logic symbol there are 3 selection lines namely S_0, S_1 & S_2 which select one of the 8 i/p's.

→ 74 x 151 IC is provides with active low enable i/p.

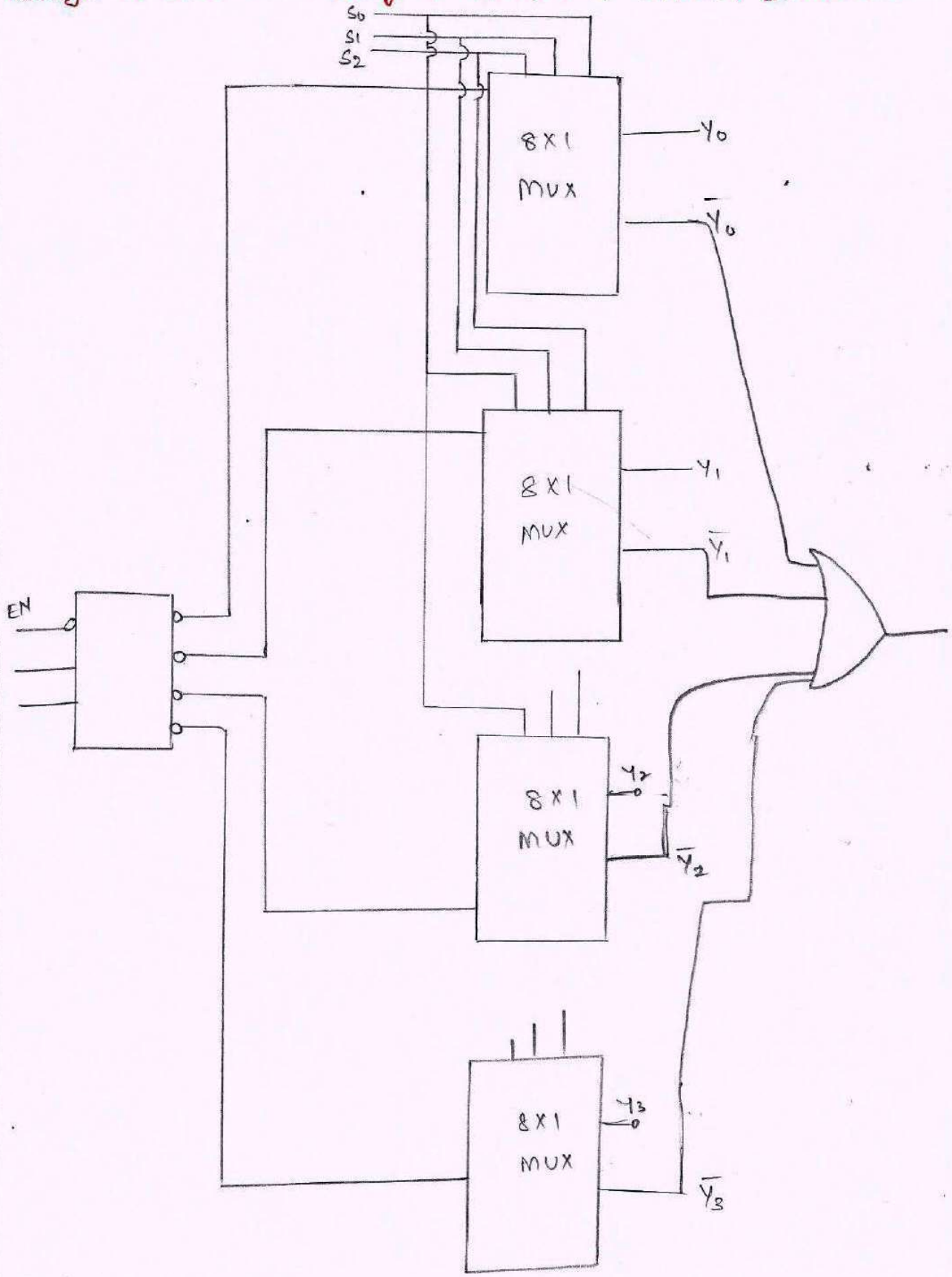
→ The truth table of an 74 x 151 is shown in below.

Enable	Selection i/p's			o/p's		
	S_0	S_1	S_2	Y	$\bar{Y}$	
1	X	X	X	0	1	
0	0	0	0	D_0	$\bar{D}_0$	
0	0	0	1	D_1	$\bar{D}_1$	
0	0	1	0	D_2	$\bar{D}_2$	
0	0	1	1	D_3	$\bar{D}_3$	

Enable	S_0	S_1	S_2	Y	$\bar{Y}$
0	1	0	0	D_4	$\bar{D}_4$
0	1	0	1	D_5	$\bar{D}_5$
0	1	1	0	D_6	$\bar{D}_6$
0	1	1	1	D_7	$\bar{D}_7$



Design a 32x1 mux using 4 74x151 & 74x139 Decoder :-



Parity Circuits :-

Parity circuit included parity bits. Parity bits are 2 types. i.e., even parity bit & odd parity bit.

→ Parity bit is used for the purpose of detecting errors during transmission of binary information.

→ A parity bit is an extra bit included with a binary message signal.

→ The message including the parity bit is transmitted & then checked at the receiving end.

→ The ckt that generates the parity bit in the transmitter is called as "Parity generator" and the circuit that checks the parity bit in the receiver is called as "Parity checker".

→ In even parity the added parity bit will make the total number of ones an even amount.

→ In odd parity the added parity bit will make the total no. of ones an odd amount.

→ Let us see the 3 bit message with even and odd parity bit.

A	B	C	odd parity	Even Parity
0	0	0	1	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

→ Let us design the even parity generator.

K-map Simplification:

A \ BC	00	01	11	10
0		1		1
1	1		1	X

$$P = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

$$= \bar{A}(\bar{B}C + B\bar{C}) + A(\bar{B}\bar{C} + BC)$$

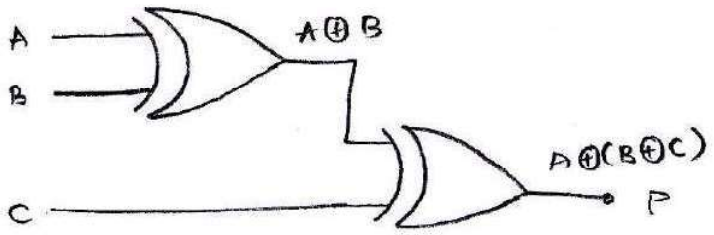
$$= \bar{A}(B \oplus C) + A(B \odot C)$$

$$= \bar{A}(B \oplus C) + A\overline{(B \oplus C)}$$

$\bar{A}B + A\bar{B}$

$A \oplus B \oplus C$

$$\therefore P = A \oplus (B \oplus C)$$



→ The parity checker circuit checks for possible errors in the transmission. Since the information was transmitted with even parity, the 4 bits received at the receiving end.

→ The o/p of parity checker is denoted by parity error check (PEC).

~~The below table shows the~~

~~the o/p of f~~

→ It will be equal to 1 if an error occurs i.e., if the 4 bits received at the receiver section.

→ The below table shows the truth table for the even parity checker.

A	B	C	D	Parity Error Checker
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

AB \ CD	00	01	11	10
00		1		1
01	1		1	
11		1		1
10	1		1	

00
01
10
11
$\bar{A}B + A\bar{B}$
10 + 01 = 0
11 + 00 = 1
00 + 11 = 1
01 + 10 = 0

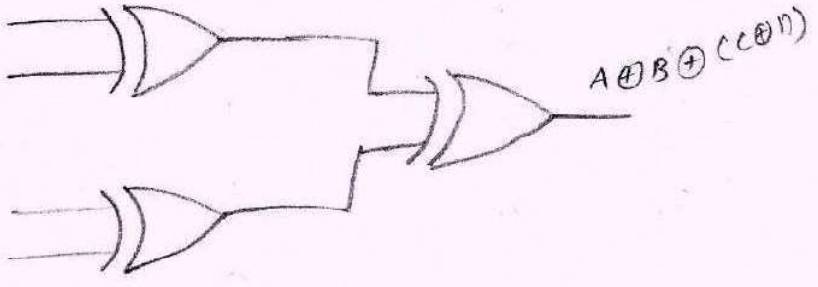
$P = \bar{A}\bar{B}\bar{C}D + \bar{A}BC\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BCD + AB\bar{C}D + ABC\bar{D} + A\bar{B}C\bar{D} + A\bar{B}CD$

$= \bar{A}\bar{B}(\bar{C}D + C\bar{D}) + \bar{A}\bar{B}(\bar{C}\bar{D} + CD) + AB(\bar{C}D + C\bar{D}) + A\bar{B}(\bar{C}\bar{D} + CD)$

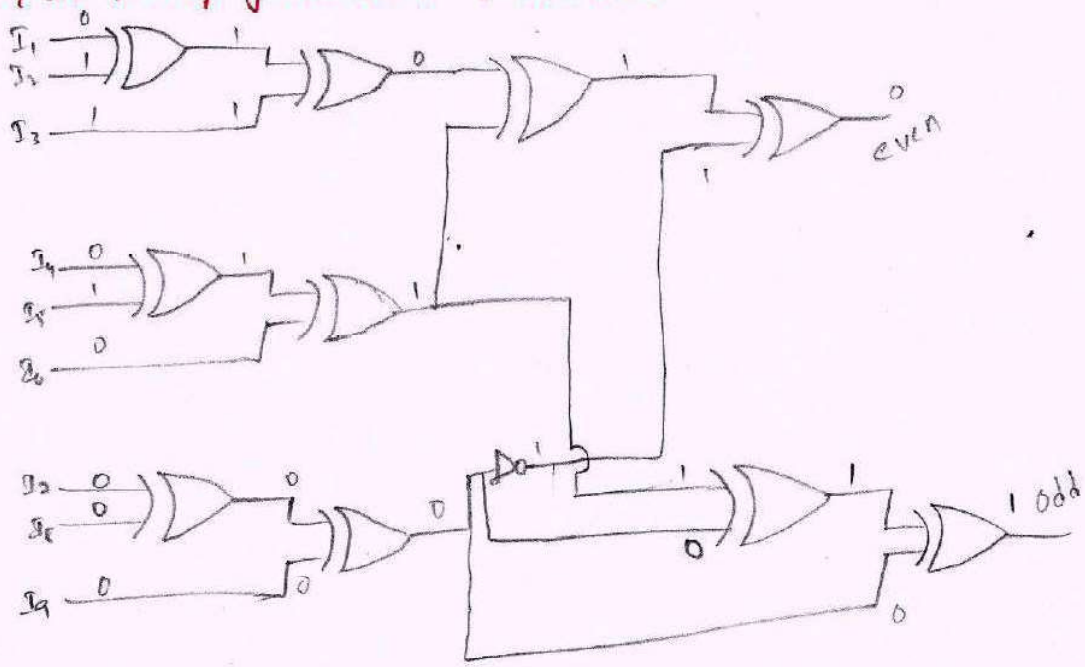
$= \bar{A}\bar{B}(C \oplus D) + \bar{A}\bar{B}(C \odot D) + AB(C \oplus D) + A\bar{B}(C \odot D)$

$= \bar{A}\bar{B}(C \oplus D) + \bar{A}B(C \odot D) + AB(C \oplus D) + A\bar{B}(C \odot D)$

$(C \oplus D)(A \oplus B) +$



9 bit Parity generator: (74 x 280)



→ 74 x 280 is a 9 bit parity generator or checker commonly used to detect errors in high speed data transmission.

→ Both even & odd parity o/p's are available for generating the 9 bits.

→ The even parity bit o/p is high when an even number of data i/p's are high.

→ The odd parity o/p bit is high when an odd no. of data i/p are high.